DevOrSysAdminScripts

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

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Code lines: 9378 total lines, 8468 not empty lines, 910 empty lines.

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1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spell_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_listing.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
```

```
│   ├── licenses_notes.txt
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   └── spampty_data.txt
├── README.html
├── README.md
├── README.md.tpl
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 64 files

[ 12K Mar  3 21:00]  ./
├── [4.0K Mar  3 21:00]  build_and_checks_dependencies/
│   ├── [3.8K Mar  2 18:25]  build_and_checks.exec.sh.tpl*
│   ├── [3.8K Mar  2 19:19]  build_md_from_printable_md.exec.sh*
│   ├── [2.8K Mar  3 08:41]  check_shell_scripts_beginnings.libr.sh
│   ├── [4.7K Mar  3 08:41]  check_shell_scripts_indentation.libr.sh
│   ├── [3.1K Mar  3 10:22]  check_URLs.libr.sh
│   ├── [ 29K Mar  3 20:54]  common_build_and_checks.exec.sh*
│   ├── [ 24K Mar  3 18:32]  common_build_and_checks.exec.sh.tpl
│   ├── [9.0K Mar  3 16:14]  common_options.libr.sh
│   ├── [4.8K Mar  3 19:09]  comparisons.libr.sh
│   ├── [ 18K Mar  3 19:03]  create_PDF.exec.sh*
│   ├── [3.3K Feb 19 18:48]  generate_from_template.libr.sh
│   ├── [1.9K Feb 19 17:18]  get_common_text_glob_patterns.libr.sh
│   ├── [2.4K Mar  3 10:22]  grammar_and_spell_check.libr.sh
│   ├── [4.0K Mar  3 19:13]  licenses_templates/
│   │   ├── [ 13K Mar  3 19:13]  build_licenses_templates.exec.sh*
```

```
│   │   ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│   │   └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│   ├── [4.8K Mar  3 18:49]  lines_counts.libr.sh
│   ├── [ 10K Mar  3 20:10]  lines_filters.libr.sh
│   ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│   ├── [1.8K Feb 26 16:58]  overwrite_if_not_equal.libr.sh
│   ├── [2.3K Mar  3 10:30]  python_black_complement.libr.sh
│   ├── [2.6K Mar  3 10:22]  python_isort_complement.libr.sh
│   ├── [2.0K Mar  3 17:16]  repository_data.libr.sh
│   ├── [1.6K Feb  3 17:57]  split_score.exec.php
│   ├── [1.9K Feb  3 20:11]  split_score.libr.php
│   ├── [ 13K Mar  3 20:35]  strings_functions.libr.sh
│   ├── [3.4K Mar  3 10:22]  too_long_code_lines.libr.sh
│   ├── [4.2K Mar  3 17:18]  update_common_build_and_checks.exec.sh*
│   └── [5.0K Mar  3 20:41]  update_or_check_files_names_listing.exec.sh*
├── [4.0K Mar  3 20:54]  build_and_checks.exec.sh*
├── [4.0K Mar  3 21:00]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [ 16K Mar  2 12:57]  DevOrSysAdminScripts.html.tpl
│   ├── [ 29K Mar  3 20:54]  DevOrSysAdminScripts.tex
│   ├── [8.5K Mar  2 13:01]  DevOrSysAdminScripts.tex.tpl
│   ├── [ 13K Mar  3 20:54]  files_listing.tex.tpl
│   ├── [4.1K Mar  3 20:39]  files_names_listing.txt
│   ├── [  24 Jan 15 22:26]  .gitignore
│   ├── [4.9K Mar  2 12:48]  licenses_notes.txt
│   ├── [1001 Feb 26 15:37]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.4K Mar  3 20:47]  repository_data.txt
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 167 Feb 19 18:39]  shellcheck.ini
│   └── [3.1K Mar  3 18:56]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [367K Mar  3 20:54]  DevOrSysAdminScripts.html
├── [333K Mar  3 20:54]  DevOrSysAdminScripts.pdf
├── [4.0K Mar  3 20:51]  git_helpers/
│   ├── [3.8K Mar  3 20:51]  git_checks.libr.sh
│   ├── [1.8K Feb 27 20:20]  post-commit*
│   ├── [ 13K Mar  3 10:50]  post_commit_renaming_comment.libr.sh
│   └── [1.4K Feb 19 18:42]  pre-commit*
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [4.0K Mar  3 19:56]  miscellaneous/
```

```
│   └── [  94 Mar  3 19:56]  spampty_data.txt
├── [7.4K Mar  1 22:30]  README.html
├── [3.7K Mar  1 22:30]  README.md
├── [3.7K Mar  1 22:29]  README.md.tpl
├── [9.1K Feb 26 15:05]  unicode.libr.php
├── [1.4K Feb  3 14:19]  upgrade_build_and_checks.exec.sh*
└── [1.4K Feb 19 14:41]  wget_sha512.libr.sh

6 directories, 64 files
```

2 Listing of files

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;

- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
`https://www.gnu.org/software/findutils/`,
`https://savannah.gnu.org/projects/findutils/`.

grep (GNU grep) can be found at the following URLs:
`https://www.gnu.org/software/grep/`,
`https://savannah.gnu.org/projects/grep/`.

sed (GNU sed) can be found at the following URLs:
`https://www.gnu.org/software/sed/`,
`https://savannah.gnu.org/projects/sed/`.

ShellCheck can be found at the following URLs:
`https://www.shellcheck.net/`,
`https://github.com/koalaman/shellcheck`.

GNU Wget can be found at the following URLs:
`https://www.gnu.org/software/wget/`,
`https://savannah.gnu.org/projects/wget/`.

more of util-linux can be found at the following URLs:
`https://www.kernel.org/pub/linux/utils/util-linux/`,
`https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/`,
`https://github.com/util-linux/util-linux` official?

pandoc can be found at the following URLs:
`https://pandoc.org/`,
`https://github.com/jgm/pandoc`.

pdftex/pdflatex can be found at the following URLs:
`https://www.tug.org/applications/pdftex/`,
`https://tug.org/texlive/svn/`.

tree can be found at the following URLs:
`https://oldmanprogrammer.net/source.php?dir=projects/tree`,

https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,

https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

If you see something that I missed regarding my dependencies, please tell me/email me :).

2.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
```

```
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    {
      echo "temp/"
      echo ""
    } > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  @sha512_common_build_and_checks.exec.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed_point_build* ]]; then
  echo "--fixed_point_build"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
```

```
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed_point_build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.2 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        echo "Searching md file: ${LFBFL_base_name}"
      fi
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
```

```
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]];
  then
    sed -Ez -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    echo "No file ${LFBFL_base_name}.md.tpl"
  fi

  pandoc --from markdown --to html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    -o "${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed -i -E -e 's~(  <meta .*) />~\1>~g'\
            -e 's~<html .*>~<html lang="en">~'\
            -e '/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
             -e "${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}
```

```
build_md_from_printable_md "$@"
```

2.3 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # declare -r LFBFL_file_path=$(basename "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head -n 1 "$1") <(echo '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
```

```
  # ------------------------------------------------------------------
  # diff <(head -n 1 "$1") <(echo '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head -n 1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || echo "$1:File has wrong shell script beginning."
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  # shopt -s globstar
  local LFBFL_file_path
  local LFBFL_head

  # for LFBFL_file_path in **/*.sh; do
  find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && echo "${LFBFL_file_path}:Checking shell script beginning."
    LFBFL_head=$(head -n 1 "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || echo "${LFBFL_file_path}:File has wrong shell script beginning."
  done
}
```

2.4 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
#!/usr/bin/env bash
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
  shopt -s lastpipe

  declare -i LFBFL_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="workeduntilsomeasshole"
  # created a command named "workeduntilsomeasshole"
  # and called it without neither indentation, nor parameters
  # in a .sh script XD.
  # Manual justification of the lines L-4 and L-3 was a serendipity.
  # I just tried a joke and was lucky ;) :).
  local LFBFL_line_end
  local LFBFL_previous_line_end
  declare -i LFBFL_line_length
  declare -i LFBFL_line_end_length
  declare -i LFBFL_line_start_length
  local LFBFL_spacing
  declare -i LFBFL_offset
  local LFBFL_substring1
  local LFBFL_substring2
  declare -r LFBFL_pattern='^ *-e '
```

```
# This grep would have been enough without some sed commands.
# shellcheck disable=SC2312
grep -EHn -B 1 '^(  )* ([^ ]|$)' "$1"\
  | while read -r LFBFL_some_line;
do
  if [[ "${LFBFL_some_line}" == "--" ]]; then
    # When switching between code fragments, we reinit.
    LFBFL_previous_line="workeduntilsomeasshole"
    continue
  fi
  # Note how it's funny that this loop always work because we check
  # shell scripts beginning in another script and thus we always
  # have a line before.
  if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]];
  then
    LFBFL_previous_line="${LFBFL_some_line}"
    continue
  fi
  # Remove the name of the file and then the line number
  LFBFL_line_end=${LFBFL_some_line#*:}
  LFBFL_line_end=${LFBFL_line_end#*:}
  # But for the previous line '-' or ':' is used instead of ':'
  # and filenames with '-' are much more frequent than with ':'...
  LFBFL_line_length=${#LFBFL_some_line}
  LFBFL_line_end_length=${#LFBFL_line_end}
  LFBFL_line_start_length=$((
    LFBFL_line_length-LFBFL_line_end_length
  ))
  LFBFL_offset=$((LFBFL_line_start_length-2))
  # Checking if increasing number of digits in the line numbers
  # (99 -> 100).
  LFBFL_substring1=${LFBFL_previous_line:${LFBFL_offset}:1}
  # Nice puzzle below :) Look for comments above :)
  case ${LFBFL_substring1} in
    -) ((LFBFL_offset+=1));;
    :) ((LFBFL_offset+=1));;
    [0-9]) ((LFBFL_offset+=2));;
    *) echo ':-(';;
  esac
  LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_offset}}
  if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
    # Line starts with spacing and -e
    # Keep max spaces
    LFBFL_spacing=${LFBFL_line_end%%-*}
    LFBFL_offset=${#LFBFL_spacing}
    LFBFL_substring1=${LFBFL_line_end:${LFBFL_offset}:3}
```

```
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        echo "${LFBFL_some_line}"
        LFBFL_file_with_error=1
      fi
    else
      echo "${LFBFL_some_line}"
      LFBFL_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_file_with_error -eq 1 ]]; then
    echo "$1:File has some lines with odd number of spaces."
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  # shopt -s globstar
  local LFBFL_file_path

  # for LFBFL_file_path in **/*.sh; do
  find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && echo "${LFBFL_file_path}:Checking shell script indentation."
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

2.5 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return
```

```
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )

  local LFBFL_pattern
  local LFBFL_file_name
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 0 ]]\
      || echo "Iterating on pattern: ${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H 'http:'\
      | grep -v "| xargs grep -H 'htt"\
      | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"
    # Last grep just above will remove false positives from
    # substitutions that fit on one line.
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find\
      | while read -r LFBFL_file_name;
    do
      [[ -f "${LFBFL_file_name}" ]] || continue
      LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      [[ LFBFL_i_verbose -eq 0 ]]\
        || echo "Handling the file: ${LFBFL_file_name}"
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=\
${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep -q "${LFBFL_substitution}" "${LFBFL_file_name}"; then
          sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            "${LFBFL_file_name}"
        fi
      done
    done
  done
}
```

2.6 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
```

```
45  if [[ "$*" == *--verbose* ]]; then
46    echo "$0 $*"
47    LFBFL_verbose="--verbose"
48    LFBFL_i_verbose=1
49  fi
50  readonly LFBFL_verbose
51  readonly LFBFL_i_verbose
52  declare -ar LFBFL_some_common_options=(
53    "${LFBFL_verbose}"
54    # "$--work-directory={LFBFL_work_directory}" doesn't work
55    "${LFBFL_work_directory_option}"
56  )
57
58  source ./wget_sha512.libr.sh
59
60  LFBFL_subdir="build_and_checks_dependencies"
61  local LFBFL_file_name
62  local LFBFL_URL
63  local LFBFL_file_path
64
65  LFBFL_file_name="build_md_from_printable_md.exec.sh"
66  # LFBFL_script_download_URL
67  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
68  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
69  local LFBFL_correct_sha512
70  LFBFL_correct_sha512='debb2bdda6f77ef82be5c17a547077d5c3db2bea62f92'
71  LFBFL_correct_sha512+='6b092815c4d7a2dbd9b8461923bff8e38199d776a71b'
72  LFBFL_correct_sha512+='c8d417fa396a556898c3d783681afb8c235922f'
73  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
74    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
75  chmod +x "./${LFBFL_file_path}"
76
77  LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
78  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
79  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
80  local LFBFL_correct_sha512
81  LFBFL_correct_sha512='13346ffc2efcff3d0628b2a28855d1cd752f2ad3882fe'
82  LFBFL_correct_sha512+='9aa3dca9c50c96e0b2780b5107d9bb7cf90eb885a80b'
83  LFBFL_correct_sha512+='6ed84ea329f6d7c06ba0e8c368e78ed9ecfe2ac'
84  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
85    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
86
87  LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
88  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
89  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
90  local LFBFL_correct_sha512
```

```
LFBFL_correct_sha512='fe468954776cc2ddbde072daa41e01181cfa7d066fc2a'
LFBFL_correct_sha512+='117647331d01b43bf04809f239bc2733ca987bd8433a'
LFBFL_correct_sha512+='aabc7e19241f3f2481268eac77a051e940524b2'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='32ff1c210a5d6e6589ec7a3a692eca090759bf9692756'
LFBFL_correct_sha512+='09dda93607349432d233674c1c79945ed92b79287c5e'
LFBFL_correct_sha512+='d044ce240b872e35462edebb60cc9f27f218ba3'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='1c5b17cbbeb7d3018f4d1bf1ce2d4c7452ce3fa8557c2'
LFBFL_correct_sha512+='a2391c448181b51e336c03fafdb1c51b9d7168837773'
LFBFL_correct_sha512+='fe15a1235ada2337efae17e54e71e99ec313760'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='445d448b0077ac8ff660bcdb8b5e3fb391a08dec1832d'
LFBFL_correct_sha512+='1f6633807806d5b145aa6fda4c7835a6510988ce3c37'
LFBFL_correct_sha512+='439b1fca19b27cb4757659cd1c204ed8aeb071f'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='e20c8bc2668914072604582043 2bcf5adc8422c69dbb9'
LFBFL_correct_sha512+='9a0b4a38bc82030c4151744d679ff372e30aa2f8a2d8'
LFBFL_correct_sha512+='57bda2a0e89d32b77ddf223c0a427007c21cd96'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"
```

```
LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='5779c3d2c64d0bb403525c410e37b0b06fcf1ef97d503'
LFBFL_correct_sha512+='21fd27345545a8ebea020b2d3d809ac5c1bc66f9cebe'
LFBFL_correct_sha512+='a8e8ea63aa539eaa179cf09a3860239c96df8e9'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='4e0915efcc9a4088b1f79c9c7efbf7da5d68e8e1b21ff'
LFBFL_correct_sha512+='feb4746c9d468be80affa30babf35938e15f9161db3e'
LFBFL_correct_sha512+='3f6d95c913b86638563d7bd2995dc58a26f8e3a'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="grammar_and_spell_check.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='6deeaae24bc1f67bfe93acbf477d8ce15311edbb6b664'
LFBFL_correct_sha512+='68052a4e29e326c33977272f4f8a6ede39cd1065f012'
LFBFL_correct_sha512+='c3171f253c230c59fc01c95f8a288eb90cb78ca'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

# /licenses_templates/ ---------------------------------------------
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='34c02434a90f63330b150ac55693b9a82baa9208a539c'
LFBFL_correct_sha512+='c6eec4cd48db32b28b32d42d6b089d148781dbaf4d49'
LFBFL_correct_sha512+='52b7b1ffd0efc8d1badbd90478d968725388cc0'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
```

```
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='809597a887cfd33d0556c3353ebd3d7cfac175cbd1d63'
LFBFL_correct_sha512+='c4181554009d81bf35b85ab068405b009f04c0a37107'
LFBFL_correct_sha512+='24db5af77178dbdbdfa18586cd5a61849f22821'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='ff7968ddd58944c291b7765b0d4cb3f9acc7ac3a94610'
LFBFL_correct_sha512+='b768c3983ff93ffc78a27893e2d58812b12f8efac2b1'
```

```
  LFBFL_correct_sha512+='40b7927987fc0cbb4de6cbf07202774adf7da87'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="lines_maps.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
  LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
  LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="overwrite_if_not_equal.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='2781a564d264bae84c787321481dc83ae43a1ce324ee6'
  LFBFL_correct_sha512+='d5ceb73d5a98607fad5fe9ff8d03979b898f7cd7c712'
  LFBFL_correct_sha512+='8215d26e76874d56aa67fb63fa9de776617d24a'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_black_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='ce29513479932cc7c2f8d79fc10ef2f3803b578c18106'
  LFBFL_correct_sha512+='9d5ba6462734acfe9907240f1c0b0fa1fecae672e722'
  LFBFL_correct_sha512+='e72c1bdf94077693cb86af1bbc34f18fb4400fa'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_isort_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='c913be48fdd63c3c518ce342860adfeb2698866548905'
  LFBFL_correct_sha512+='6aa3e3299fc81f0c418e30d5d6ead46804e9b426a39f'
  LFBFL_correct_sha512+='22fffcb1c89f7c2a380915b66acaa54dba6af1d'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="repository_data.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='236fe88b20c3e900e840651e489f84534036262ae048b'
LFBFL_correct_sha512+='fcfdd494c98cbdba17d634e4c842c5b37866bc0332ae'
LFBFL_correct_sha512+='e322e8fc140e7e8e0f1b2209cf839b6d56066a7'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.exec.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2f7ff107a43716109b1acbd7d89870d05577af00a981a'
LFBFL_correct_sha512+='05f162a054c96387fb99544f8225c333b1212f74326c'
LFBFL_correct_sha512+='f9ec6196e796f6257dfbba9347ac8c57b4f09db'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.libr.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='1413898a4fb4f31ca17993fd6110f7e1f32adf31c22c4'
LFBFL_correct_sha512+='7fa46a31d3c1511a3112a611e04fac19bb838d4476d9'
LFBFL_correct_sha512+='ecf77a28dd48d3d2732df7392c50e38262852d7'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='4abbc3b1ad8ccf7f73eb834cd66f4781af623546ec5e6'
LFBFL_correct_sha512+='cb1eec69c9520fd9578538ddedbde624416ec3572b95'
LFBFL_correct_sha512+='7a183c5feea398e65cf9b1a9b47f195960151d6'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='fb5571eead378f7de8add4a277a71450f82d095990219'
LFBFL_correct_sha512+='6ed559bfd2896511d406983d19b7816c62c2bb9ebc6c'
LFBFL_correct_sha512+='7ae6b877c74b466b19904ed0e12282ae0a5475c'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
```

```
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='f4f91e21cbe4084dd85bd0a1787519834147acdb4c887'
  LFBFL_correct_sha512+='dc8460ef21c5434e3e37d09dbe14c945c6f6cac7cc9f'
  LFBFL_correct_sha512+='b1e16e288f97498d7a89a2d92744d658524b4a0'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  # shellcheck source=check_shell_scripts_beginnings.libr.sh
  source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
  # shellcheck source=check_shell_scripts_indentation.libr.sh
  source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
  # shellcheck source=check_URLs.libr.sh
  source "./${LFBFL_subdir}/check_URLs.libr.sh"
  # shellcheck source=common_options.libr.sh
  source "./${LFBFL_subdir}/common_options.libr.sh"
  # shellcheck source=comparisons.libr.sh
  source "./${LFBFL_subdir}/comparisons.libr.sh"
  # shellcheck source=generate_from_template.libr.sh
  source "./${LFBFL_subdir}/generate_from_template.libr.sh"
  # shellcheck source=get_common_text_glob_patterns.libr.sh
  source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
  # shellcheck source=grammar_and_spell_check.libr.sh
  source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
  # shellcheck source=lines_counts.libr.sh
  source "./${LFBFL_subdir}/lines_counts.libr.sh"
  # shellcheck source=lines_filters.libr.sh
  source "./${LFBFL_subdir}/lines_filters.libr.sh"
  # shellcheck source=lines_maps.libr.sh
  source "./${LFBFL_subdir}/lines_maps.libr.sh"
  # shellcheck source=overwrite_if_not_equal.libr.sh
  source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
  # shellcheck source=python_black_complement.libr.sh
  source "./${LFBFL_subdir}/python_black_complement.libr.sh"
  # shellcheck source=python_isort_complement.libr.sh
  source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
  # shellcheck source=strings_functions.libr.sh
  source "./${LFBFL_subdir}/strings_functions.libr.sh"
  # shellcheck source=too_long_code_lines.libr.sh
  source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

```

```
  enhanced_set_bash_option extglob
  # shellcheck source=repository_data.libr.sh
  source "./${LFBFL_subdir}/repository_data.libr.sh"
  # shellcheck disable=SC2154,SC2309
  [[ enhanced_set_bash_option_extglob_result -eq 0 ]]\
    && enhanced_unset_bash_option extglob

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  echo "Building license headers"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  echo "Building README.md"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  echo "Building other MarkDown files"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  find "${LFBFL_work_directory}" -name "*.md.tpl"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    echo "Found template ${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
    LFBFL_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
      "--work-directory=${LFBFL_some_directory}"\
      "--base-name=${LFBFL_file_name}"\
      "${LFBFL_verbose}"
  done

  declare -i LFBFL_will_popd
  pushd_to_work_directory
  LFBFL_will_popd=$?
  can_continue_after_enhanced_pushd || return

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

```

```
  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_upgrade_venvs_ts
  declare -i LFBFL_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      LFBFL_upgrade_venvs=1
    fi
  else
    LFBFL_upgrade_venvs=1
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_upgrade_venvs=0
      fi
    fi
  fi

  echo "Running shellcheck"
  find . -name "*.sh"\
    | relevant_find\
    | while read -r LFBFL_file_path;
```

```
  do
    shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
      "${LFBFL_file_path}"
  done

  echo "---Python---"
  echo "Running isort"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  echo "Running python_isort_complement"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  echo "Running black"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
```

```
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  echo "Running python_black_complement"
  python_black_complement "${LFBFL_some_common_options[@]}"

  echo "Probing if mypy should be runned"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_no_toml=1
  find . -name "pyproject.toml"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
      echo "Running mypy"
      LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
      mypy "${LFBFL_directory_path}"
    fi
    LFBFL_no_toml=0
  done
  if [[ LFBFL_no_toml -eq 1 ]]; then
```

```
    echo "Running mypy"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  echo "Running bandit"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini build_and_checks_variables/bandit.ini\
    -b build_and_checks_variables/bandit_baseline.json\
    -r .
  # Saving new baseline in temp if necessary.
  bandit --ini build_and_checks_variables/bandit.ini\
    -f json -o build_and_checks_variables/temp/bandit_baseline.json\
    -r .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  echo "Running pylint"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_pylint_venv}/bin/activate"
```

```
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile build_and_checks_variables/pylintrc.toml\
    --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  echo "Running ruff"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config build_and_checks_variables/ruff.toml
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  echo "---Python end---"

  echo "---PHP---"
  echo "Running PHPMD"
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  local LFBFL_phpmd_baseline="build_and_checks_variables/"
  LFBFL_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_phpmd_baseline

  local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

```

```
643   local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
644   LFBFL_phpmd_rulesets+="design,naming,unusedcode"
645   readonly LFBFL_phpmd_rulesets
646
647   phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
648     . text "${LFBFL_phpmd_rulesets}"
649   # Saving new baseline in temp if necessary.
650   rm "${LFBFL_temp_phpmd_baseline}"
651   phpmd --color --generate-baseline\
652     --baseline-file "${LFBFL_temp_phpmd_baseline}"\
653     . text "${LFBFL_phpmd_rulesets}"
654   echo "---PHP end---"
655
656   echo "---JS---"
657   LFBFL_npm_lint_directories=""
658   grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
659     --result-variable-prefix="LFBFL_"
660   if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
661     echo "Running ESLint"
662     local LFBFL_JS_directory
663     echo "${LFBFL_npm_lint_directories}"\
664       | sed -e 's/\\n/\n/g'\
665       | while read -r LFBFL_JS_directory;
666     do
667       (cd "${LFBFL_JS_directory}" && npm run lint)
668     done
669   fi
670   echo "---JS end---"
671
672   echo "Analyzing too long lines"
673   # shellcheck disable=SC2248
674   too_long_code_lines "${LFBFL_some_common_options[@]}" \
675     --max-line-length=${LFBFL_max_line_length}
676
677   echo "Analyzing shell scripts beginnings"
678   check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
679     | relevant_grep
680
681   echo "Analyzing shell scripts indentation"
682   check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
683     | relevant_grep
684
685   echo "Analyzing URLs"
686   check_URLs "${LFBFL_some_common_options[@]}" \
687     | relevant_grep
688
```

```
  echo "Analyzing strange characters: hover over in doubt"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_will_popd -eq 0 ]] && popd_from_work_directory

  echo "Checking listed files"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  echo "Creating the PDF file of the listing of the source code"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  echo "Running Nu W3C HTML CSS and SVG validator"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
```

```
    ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      echo $'/!\\ATTENTION: Nu W3C validator is too old./!\\'
      echo "Go get the latest version here:"
      echo "https://github.com/validator/validator/releases"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    declare -a LFBFL_files_for_Nu2
    mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
    readonly LFBFL_files_for_Nu2
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose "${LFBFL_files_for_Nu2[@]}"
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.7 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
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# of the GNU General Public License
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# either version 3 of the License,
# or (at your option) any later version.
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# the GNU General Public License
```

```
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  declare -ar LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    # "$--work-directory={LFBFL_work_directory}" doesn't work
    "${LFBFL_work_directory_option}"
  )

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  local LFBFL_file_name
  local LFBFL_URL
  local LFBFL_file_path

  LFBFL_file_name="build_md_from_printable_md.exec.sh"
```

```
# LFBFL_script_download_URL
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_md_from_printable_md.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_beginnings.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_indentation.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_URLs.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_common_options.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_comparisons.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
@sha512_create_PDF.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_generate_from_template.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_get_common_text_glob_patterns.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="grammar_and_spell_check.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_grammar_and_spell_check.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

# /licenses_templates/ ---------------------------------------------
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_build_licenses_templates.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_AGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_GPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_maps.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_black_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
```

```
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="python_isort_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_python_isort_complement.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="repository_data.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_repository_data.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.exec.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_split_score.exec.php@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.libr.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_split_score.libr.php@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="strings_functions.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_strings_functions.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="too_long_code_lines.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_too_long_code_lines.libr.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_update_or_check_files_names_listing.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spell_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

```

```
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  echo "Building license headers"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  echo "Building README.md"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  echo "Building other MarkDown files"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  find "${LFBFL_work_directory}" -name "*.md.tpl"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    echo "Found template ${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
    LFBFL_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
      "--work-directory=${LFBFL_some_directory}"\
      "--base-name=${LFBFL_file_name}"\
      "${LFBFL_verbose}"
  done

  declare -i LFBFL_will_popd
  pushd_to_work_directory
  LFBFL_will_popd=$?
  can_continue_after_enhanced_pushd || return

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
```

```
342   declare -i LFBFL_upgrade_venvs_ts
343   declare -i LFBFL_current_ts
344   local LFBFL_upgrade_venvs_answer
345
346   local LFBFL_upgrade_venvs_time_interval_in_seconds=""
347   get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
348     "${LFBFL_some_common_options[@]}"
349
350   if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
351     LFBFL_upgrade_venvs_ts=$(
352       stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
353     )
354     LFBFL_current_ts=$(date +%s)
355     ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
356     ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
357     if [[ LFBFL_current_ts -gt 0 ]]; then
358       LFBFL_upgrade_venvs=1
359     fi
360   else
361     LFBFL_upgrade_venvs=1
362   fi
363   if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
364     local LFBFL2_upgrade_venvs=""
365     grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
366       --result-variable-prefix="LFBFL2_"
367     if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
368       read -r -n 1 -t 10\
369         -p "Upgrade venvs and composer global? [Y/n]"\
370         LFBFL_upgrade_venvs_answer
371       if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
372         LFBFL_upgrade_venvs=0
373       fi
374     fi
375   fi
376
377   echo "Running shellcheck"
378   find . -name "*.sh"\
379     | relevant_find\
380     | while read -r LFBFL_file_path;
381   do
382     shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
383       "${LFBFL_file_path}"
384   done
385
386   echo "---Python---"
387   echo "Running isort"
```

```
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  echo "Running python_isort_complement"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  echo "Running black"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
```

```
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  echo "Running python_black_complement"
  python_black_complement "${LFBFL_some_common_options[@]}"

  echo "Probing if mypy should be runned"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_no_toml=1
  find . -name "pyproject.toml"\
    | relevant_find\
    | while read -r LFBFL_file_path;
  do
    if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
      echo "Running mypy"
      LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
      mypy "${LFBFL_directory_path}"
    fi
    LFBFL_no_toml=0
  done
  if [[ LFBFL_no_toml -eq 1 ]]; then
    echo "Running mypy"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

```

```
echo "Running bandit"
LFBFL_bandit_venv=""
grep_variable "${LFBFL_data_file_name}" bandit_venv\
  --result-variable-prefix="LFBFL_"\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_bandit_venv}" ]]; then
  if command -v deactivate
  then
    deactivate
  fi
  # shellcheck disable=SC1090,SC1091
  source "${LFBFL_bandit_venv}/bin/activate"
fi
if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade bandit
fi
bandit --ini build_and_checks_variables/bandit.ini\
  -b build_and_checks_variables/bandit_baseline.json\
  -r .
# Saving new baseline in temp if necessary.
bandit --ini build_and_checks_variables/bandit.ini\
  -f json -o build_and_checks_variables/temp/bandit_baseline.json\
  -r .
if [[ -n "${LFBFL_bandit_venv}" ]]; then
  deactivate
fi

echo "Running pylint"
LFBFL_pylint_venv=""
grep_variable "${LFBFL_data_file_name}" pylint_venv\
  --result-variable-prefix="LFBFL_"\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_pylint_venv}" ]]; then
  if command -v deactivate
  then
    deactivate
  fi
  # shellcheck disable=SC1090,SC1091
  source "${LFBFL_pylint_venv}/bin/activate"
fi
if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade pylint
fi
pylint --rcfile build_and_checks_variables/pylintrc.toml\
  --recursive=y .
if [[ -n "${LFBFL_pylint_venv}" ]]; then
```

```
    deactivate
  fi

  echo "Running ruff"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config build_and_checks_variables/ruff.toml
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  echo "---Python end---"

  echo "---PHP---"
  echo "Running PHPMD"
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  local LFBFL_phpmd_baseline="build_and_checks_variables/"
  LFBFL_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_phpmd_baseline

  local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

  phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  # Saving new baseline in temp if necessary.
```

```
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file "${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  echo "---PHP end---"

  echo "---JS---"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
    --result-variable-prefix="LFBFL_"
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    echo "Running ESLint"
    local LFBFL_JS_directory
    echo "${LFBFL_npm_lint_directories}"\
      | sed -e 's/\\n/\n/g'\
      | while read -r LFBFL_JS_directory;
    do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  echo "---JS end---"

  echo "Analyzing too long lines"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options[@]}" \
    --max-line-length=${LFBFL_max_line_length}

  echo "Analyzing shell scripts beginnings"
  check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  echo "Analyzing shell scripts indentation"
  check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  echo "Analyzing URLs"
  check_URLs "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  echo "Analyzing strange characters: hover over in doubt"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_will_popd -eq 0 ]] && popd_from_work_directory
```

```
618
619   echo "Checking listed files"
620   "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
621     "${LFBFL_some_common_options[@]}"
622
623   echo "Creating the PDF file of the listing of the source code"
624   "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
625
626   pushd_to_work_directory\
627     && trap 'popd_from_work_directory' RETURN
628   can_continue_after_enhanced_pushd || return
629
630   if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
631     ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
632   fi
633
634   # ------------------------------------------------------------------
635   # It is at the end that all the HTML files are generated and that
636   # we can check there are no mistakes.
637   echo "Running Nu W3C HTML CSS and SVG validator"
638   local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
639   grep_variable "${LFBFL_data_file_name}"\
640     upgrade_vnu_jar_time_interval_in_seconds\
641     --result-variable-prefix="LFBFL_"
642
643   if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
644   then
645     LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
646   fi
647
648   declare -i LFBFL_vnu_jar_ts
649   local LFBFL_vnu_jar_path=""
650   grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
651     --result-variable-prefix="LFBFL_"\
652     --replace-line-returns-by=""
653   if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
654     LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
655     LFBFL_current_ts=$(date +%s)
656     ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
657     ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
658     if [[ LFBFL_current_ts -gt 0 ]]; then
659       echo $'/!\\ATTENTION: Nu W3C validator is too old./!\\'
660       echo "Go get the latest version here:"
661       echo "https://github.com/validator/validator/releases"
662     fi
663     declare -r LFBFL_files_for_Nu=$(
```

```
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    declare -a LFBFL_files_for_Nu2
    mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
    readonly LFBFL_files_for_Nu2
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose "${LFBFL_files_for_Nu2[@]}"
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.8 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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get_where_was_i(){
  # Auxiliary function needed below.
```

```
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd requested: $1"
  declare -ig enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd validated1"
  is_top_dirstack_directory "$1" && {
    enhanced_pushd_result=111
    # shellcheck disable=SC2248
```

```
    return ${enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd validated2"

  pushd "$1" || {
    enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    echo "${LFBFL_where_was_i} $3$1 no such directory."
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_pushd executed"
  # shellcheck disable=SC2248
  return ${enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_popd requested: $1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    echo "${LFBFL_where_was_i} popd failed."
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && echo "enhanced_popd executed"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ enhanced_pushd_result -eq 0\
  || enhanced_pushd_result -eq 110\
  || enhanced_pushd_result -eq 111\
  ]]
```

```
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    readonly LFBFL_i_popd_result
    echo "${LFBFL_where_was_i} $*"
    LFBFL_i_verbose=1
  fi
  # shellcheck disable=SC2034
  readonly LFBFL_i_verbose
}

```

```

# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --work-directory=* ]]; then
      LFBFL_work_directory=${LFBFL_arg#--work-directory=}
      LFBFL_work_directory=${LFBFL_work_directory:-.}
      # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
      LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
      break
    fi
  done
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
}

popd_from_work_directory(){
  enhanced_popd "${LFBFL_work_directory}" 3
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
```

```
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  declare -r LFBFL_result_name="enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    echo "${LFBFL_where_was_i} $1 shell option activated."
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  declare -r LFBFL_result_name="enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set +o "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    echo "${LFBFL_where_was_i} $1 shell option unactivated."
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

```

enhanced_set_bash_option(){
  # $1=optname some bash option for shopt
  declare -r LFBFL_result_name="enhanced_set_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
  # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
  if shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -s "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    echo "${LFBFL_where_was_i} $1 bash option activated."
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  declare -r LFBFL_result_name="enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    echo "${LFBFL_where_was_i} $1 bash option unactivated."
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

2.9 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_first=1
  local LFBFL_current
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ LFBFL_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_arg}"
      LFBFL_first=0
      continue
    fi
    if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
      return 0
    fi
  done
  return 1
```

```
}

some_distinct(){
  ! equal "$@"
}

all_distinct(){
  # Returns 1 if all the arguments strings are distinct.
  declare -ir LFBFL_ij_max=$#
  declare -i LFBFL_i
  declare -i LFBFL_j
  for ((LFBFL_i=1; LFBFL_i<=LFBFL_ij_max; ++LFBFL_i)); do
    for ((LFBFL_j=1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
      [[ LFBFL_i -eq LFBFL_j ]] && continue
      [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
    done
  done
  return 1
}

all_distinct2(){
  # Returns 1 if all the arguments strings are distinct.
  local LFBFL_element
  local LFBFL_previous_element
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  # Not setting pipefail since the result would still be incorrect.
  # shellcheck disable=SC2312
  printf "%s\n" "$@" \
    | sort\
    | {
      read -r LFBFL_previous_element\
      && while read -r LFBFL_element; do
        if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]];
        then
          return 0
        fi
        LFBFL_previous_element="${LFBFL_element}"
      done
      return 1
    }
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
```

```
  # This is the fastest of all 3; but note that version 2 is faster
  # than this one if a call to sort is added to this one.
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  # Not setting pipefail since the result would still be incorrect.
  # shellcheck disable=SC2162,SC2312
  printf "%s\n" "$@" \
    | uniq -d\
    | read
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1 -r"\
    | head -1
}

min(){
  # Returns the minimum string among the arguments.
  # Renvoie la chaîne la plus petite parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ min sort a b c
  # a
```

```
142   if [[ $# -eq 0 ]]; then
143     return 1
144   fi
145   enhanced_set_shell_option pipefail\
146     && trap 'enhanced_unset_shell_option pipefail' RETURN
147   printf "%s\n" "${@:2}" \
148     | eval "$1"\
149     | head -1
150 }
151
152 is_substring(){
153   # $1=$string
154   # $2=$substring
155   declare -r LFBFL_var_1=$(
156     echo "$1"\
157     | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
158   )
159   declare -r LFBFL_var_2=$(
160     echo "$2"\
161     | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
162   )
163   local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
164   LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
165   LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
166   LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
167   readonly LFBFL_var_3
168   php -r "${LFBFL_var_3}"
169   return $?
170 }
171
172 is_subfile(){
173   # $1=$file_name
174   # $2=$sub_file_name
175   local LFBFL_var_1="\$a = file_get_contents('$1');"
176   LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
177   LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
178   LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
179   readonly LFBFL_var_1
180   php -r "${LFBFL_var_1}"
181   return $?
182 }
183
184 alias min_from_int_sort="min 'sort --numeric-sort'"
185 alias max_from_int_sort="max 'sort --numeric-sort'"
```

2.10 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
```

```
45    #   --work-directory=""
46    declare -i LFBFL_i_verbose=0
47    get_verbose_option "$@"
48    local LFBFL_work_directory=""
49    get_work_directory_option "$@"
50    pushd_to_work_directory\
51      && trap 'popd_from_work_directory' RETURN
52    can_continue_after_enhanced_pushd || return
53
54    enhanced_set_shell_option pipefail\
55      && trap 'enhanced_unset_shell_option pipefail' RETURN
56
57    local LFBFL_subdir2="build_and_checks_variables"
58    local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
59    local LFBFL_repository_name=""
60    grep_variable "${LFBFL_data_file_name}" repository_name\
61      --result-variable-prefix="LFBFL_"\
62      --replace-line-returns-by=""
63
64    local LFBFL_repository_web_url=""
65    grep_variable "${LFBFL_data_file_name}" repository_web_url\
66      --result-variable-prefix="LFBFL_"\
67      --replace-line-returns-by=""
68
69    local LFBFL_repository_git_url=""
70    grep_variable "${LFBFL_data_file_name}" repository_git_url\
71      --result-variable-prefix="LFBFL_"\
72      --replace-line-returns-by=""
73
74    local LFBFL_abstract=""
75    grep_variable "${LFBFL_data_file_name}" abstract\
76      --result-variable-prefix="LFBFL_"
77
78    local LFBFL_acknowledgments=""
79    grep_variable "${LFBFL_data_file_name}" acknowledgments\
80      --result-variable-prefix="LFBFL_"
81
82    local LFBFL_author_email=""
83    grep_variable "${LFBFL_data_file_name}" author_email\
84      --result-variable-prefix="LFBFL_"\
85      --replace-line-returns-by=""
86
87    local LFBFL_author_full_name=""
88    grep_variable "${LFBFL_data_file_name}" author_full_name\
89      --result-variable-prefix="LFBFL_"\
90      --replace-line-returns-by=" "
```

```
  local LFBFL_author_website=""
  grep_variable "${LFBFL_data_file_name}" author_website\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -r LFBFL_current_date=$(date -I"date")

  declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)

  declare -ir LFBFL_number_of_commits=$(
    git shortlog\
    | space_starting_lines\
    | wc -l
  )

  local LFBFL_redirect="/dev/stdout"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_redirect="/dev/null"
  fi
  {
    code_lines_count_all "$@"
    code_lines_count_not_empty "$@"
    code_lines_count_empty "$@"
  } &> "${LFBFL_redirect}"

  local LFBFL_number_of_lines="${code_lines_count_all_result}"
  LFBFL_number_of_lines+=" total lines,"
  LFBFL_number_of_lines+=" ${code_lines_count_not_empty_result}"
  LFBFL_number_of_lines+=" not empty lines,"
  LFBFL_number_of_lines+=" ${code_lines_count_empty_result}"
  LFBFL_number_of_lines+=" empty lines."
  readonly LFBFL_number_of_lines

  # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
  # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
  # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
  tree -a --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
```

```
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_subdir2}/temp/current_tree_light.txt"

  tree -a -DFh --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_subdir2}/temp/current_tree.txt.temp"

  local LFBFL_temp_files_listing="./${LFBFL_subdir2}/temp/"
  LFBFL_temp_files_listing+="files_listing.tex.tpl.temp"
  readonly LFBFL_temp_files_listing
  : > "${LFBFL_temp_files_listing}"
  local LFBFL_temp_files_listing2="./${LFBFL_subdir2}/temp/"
  LFBFL_temp_files_listing2+="files_listing.html.tpl.temp"
  readonly LFBFL_temp_files_listing2
  : > "${LFBFL_temp_files_listing2}"
  # HTML <li> elements, hence "lis".
  local LFBFL_temp_files_lis
  LFBFL_temp_files_lis="./${LFBFL_subdir2}/temp/files_lis.html.tpl"
  readonly LFBFL_temp_files_lis
  : > "${LFBFL_temp_files_lis}"
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} /
  declare -r LFBFL_score_command="${get_split_score_result}"
  local LFBFL_score_command_properties="${get_split_score_result2}"
  readonly LFBFL_score_command_properties
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_max_line_length} ':'
  declare -r LFBFL_score_command2="${get_split_score_result}"
  local LFBFL_score_command_properties2="${get_split_score_result2}"
  readonly LFBFL_score_command_properties2
  declare -r LFBFL_suffix='%'
  declare -i LFBFL_i=0
  local LFBFL_cleaned_path1
  local LFBFL_cleaned_path2
  local LFBFL_new_lines
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_name
  # Remove line returns here to keep lines short.
  sed -Ez 's/\\\n//Mg' "./${LFBFL_subdir2}/files_names_listing.txt"\
```

```
    | grep -v '^//'\
    | while read -r LFBFL_file_name;
  do
    echo "Listing file for tex/HTML : ${LFBFL_file_name}"
    # echo "Listing file for tex : ${LFBFL_file_name}"
    # LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    LFBFL_cleaned_path1="${LFBFL_file_name//_/\\_}"
    LFBFL_cleaned_path2=$(
      echo "${LFBFL_file_name}"\
      | sed -e 's/\//:/g' -e 's/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        "${LFBFL_score_command_properties2}"
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_name}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    {
```

```
      echo "\subsection{"
      echo "  ${LFBFL_file_name}"\
        | sed -e "s|  ${LFBFL_file_name}|${LFBFL_new_lines}|g"\
              -e 's/_/\\_/g'
      echo "}"
      echo "\label{"
      echo "  ${LFBFL_cleaned_path2}"\
        | sed -e "s|  ${LFBFL_cleaned_path2}|${LFBFL_new_lines2}|g"
      echo "}"
      echo ""
      echo "\VerbatimInput[numbers=left,xleftmargin=-5mm]{"
      echo "  ${LFBFL_file_name}"\
        | sed -e "s|  ${LFBFL_file_name}|${LFBFL_new_lines3}|g"
      echo "}"
      echo ""
      echo ""
    } >> "${LFBFL_temp_files_listing}"

    # echo "Listing file for HTML : ${LFBFL_file_name}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_name}"
    if [[ ${#LFBFL_file_name} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      echo "<h3 id=\"subsection2.${LFBFL_i}\">2.${LFBFL_i}"
      # shellcheck disable=SC2001
      echo "${LFBFL_file_name}"\
        | sed -e "s|${LFBFL_file_name}|${LFBFL_new_lines}|g"
      echo "</h3>"
      echo "<pre class=\"numbered_lines\">"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_name}"
      echo "</pre>"
      echo ""
      echo ""
    } >> "${LFBFL_temp_files_listing2}"
    {
      echo "      <li><a href=\"#subsection2.${LFBFL_i}\">"
```

```
      # shellcheck disable=SC2001
      echo "${LFBFL_file_name}"\
        | sed -e "s|${LFBFL_file_name}|${LFBFL_new_lines}|g"
      echo "      </a></li>"
    } >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "./${LFBFL_subdir2}/files_listing.tex.tpl"\
    "${LFBFL_temp_files_listing}"
  sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
  overwrite_if_not_equal\
    "./${LFBFL_subdir2}/temp/files_listing.html.tpl"\
    "${LFBFL_temp_files_listing2}"

  # We verify if some lines are beyond max_line_length characters
  # in current_tree_light.txt and current_tree.txt.
  declare -ar LFBFL_trees=(
    "current_tree_light.txt"
    "current_tree.txt.temp"
  )
  local LFBFL_tree_path
  local LFBFL_line
  local LFBFL_prefix
  declare -i LFBFL_prefix_last_position
  local LFBFL_char
  local LFBFL_line_start
  declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
  local LFBFL_file_name2
  for LFBFL_tree in "${LFBFL_trees[@]}"; do
    LFBFL_tree_path="${LFBFL_subdir2}/temp/${LFBFL_tree}"
    # shellcheck disable=SC2248
    grep '.\{'${LFBFL_overlength}'\}' "${LFBFL_tree_path}"\
      | while read -r LFBFL_line;
    do
      # echo "LFBFL_line: ${LFBFL_line}"
      LFBFL_prefix=$(
        echo "${LFBFL_line}"\
        | sed -E -e 's/(.*)——[^—]+$/\1/g'
      )
      # echo "LFBFL_prefix: ${LFBFL_prefix}"
      LFBFL_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_prefix_last_position}:1}"
      # echo "LFBFL_char: ${LFBFL_char}"
      LFBFL_prefix=$(
        echo "${LFBFL_prefix}"\
        | sed -E -e 's/[^ ]+$//g'
      )
```

```
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # echo "LFBFL_prefix: ${LFBFL_prefix}"
      LFBFL_file_name=$(
        echo "${LFBFL_line}"\
        | sed -E 's|.* (([a-zA-Z0-9\._/-]+).)$|\1|g'
      )
      # echo "LFBFL_file_name: ${LFBFL_file_name}"
      LFBFL_file_name2=$(
        echo "${LFBFL_file_name}"\
        | sed -E -e 's/\*/\\\*/g'
      )
      LFBFL_line_start=$(
        echo "${LFBFL_line}"\
        | sed -E -e "s/(.*)[ ]*${LFBFL_file_name2}/\1/g"\
                 -e 's/ *$//g'
      )
      # echo "LFBFL_line_start: ${LFBFL_line_start}"
      LFBFL_line=$(
        echo "${LFBFL_line}"\
        | sed -E -e 's/\[/\\\[/g' -e 's/\]/\\\]/g' -e 's/\*/\\\*/g'
      )
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      sed -i -e\
        "s/${LFBFL_line}/${LFBFL_line_start}\n${LFBFL_new_lines}/g"\
        "${LFBFL_tree_path}"
    done
  done

  overwrite_if_not_equal "${LFBFL_subdir2}/temp/current_tree.txt"\
    "${LFBFL_subdir2}/temp/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
```

```
"./${LFBFL_subdir2}/temp/${LFBFL_repository_name}.tex"
  declare -r LFBFL_html_path_start=\
"./${LFBFL_subdir2}/temp/${LFBFL_repository_name}.html"

  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]];
  then
    # If there is a template, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]];
  then
    # Otherwise if there is a tex file, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    echo "Neither .tex.tpl, nor .tex in ./${LFBFL_subdir2}/"
  fi

  # Same logic with repository HTML file.
  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]];
  then
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "./${LFBFL_repository_name}.html" ]]; then
    cp "./${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    echo "Neither .html.tpl in ./${LFBFL_subdir2}/, nor .html in ./"
  fi

  # shellcheck disable=SC2001
  echo "${LFBFL_abstract}"\
    | sed -e 's/\\n/\n/g'\
    > "./${LFBFL_subdir2}/temp/abstract_temp"

  # shellcheck disable=SC2001
  echo "${LFBFL_acknowledgments}"\
    | sed -e 's/\\n/\n/g'\
    > "./${LFBFL_subdir2}/temp/acknowledgments_temp"

  local LFBFL_temp_path
  LFBFL_temp_path=$(realpath "./${LFBFL_subdir2}/temp/")
  readonly LFBFL_temp_path

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
```

```
413   # Tex filling:
414   if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
415     sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
416            -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
417            -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
418            -e "s|@author_email@|${LFBFL_author_email}|g"\
419            -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
420            -e "s|@author_website@|${LFBFL_author_website}|g"\
421            -e "s|@current_date@|${LFBFL_current_date}|g"\
422            -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
423            -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
424            -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
425       "${LFBFL_tex_path_start}.1"
426
427     insert_file_at_token "${LFBFL_tex_path_start}.1" @abstract@\
428       "./${LFBFL_subdir2}/temp/abstract_temp"\
429       "${LFBFL_tex_path_start}.2"
430
431     insert_file_at_token "${LFBFL_tex_path_start}.2"\
432       @acknowledgments@\
433       "./${LFBFL_subdir2}/temp/acknowledgments_temp"\
434       "${LFBFL_tex_path_start}.3"
435
436     enhanced_pushd "${LFBFL_temp_path}" 2
437     can_continue_after_enhanced_pushd || return
438     sed -i\
439       -e '/@current_tree_light@/{r current_tree_light.txt' -e 'd}'\
440       -e '/@current_tree@/{r current_tree.txt' -e 'd}'\
441       "${LFBFL_repository_name}.tex.3"
442     if [[ enhanced_pushd_result -eq 0 ]]; then
443       enhanced_popd "${LFBFL_temp_path}" 2\
444         || return
445     fi
446
447     insert_file_at_token "${LFBFL_tex_path_start}.3"\
448       @files_listing_VerbatimInput@\
449       "./${LFBFL_subdir2}/files_listing.tex.tpl"\
450       "${LFBFL_tex_path_start}.4"
451
452     overwrite_if_not_equal\
453       "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
454       "${LFBFL_tex_path_start}.4" 1
455   fi
456
457   # HTML filling:
458   declare -i LFBFL_HTML_updated=0
```

```
459   if [[ -f "${LFBFL_html_path_start}.1" ]]; then
460     sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
461            -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
462            -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
463            -e "s|@author_email@|${LFBFL_author_email}|g"\
464            -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
465            -e "s|@author_website@|${LFBFL_author_website}|g"\
466            -e "s|@current_date@|${LFBFL_current_date}|g"\
467            -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
468            -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
469            -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
470       "${LFBFL_html_path_start}.1"
471
472     insert_file_at_token "${LFBFL_html_path_start}.1" @abstract@\
473       "./${LFBFL_subdir2}/temp/abstract_temp"\
474       "${LFBFL_html_path_start}.2"
475
476     insert_file_at_token "${LFBFL_html_path_start}.2"\
477       @acknowledgments@\
478       "./${LFBFL_subdir2}/temp/acknowledgments_temp"\
479       "${LFBFL_html_path_start}.3"
480
481     insert_file_at_token "${LFBFL_html_path_start}.3"\
482       @files_lis@ "${LFBFL_temp_files_lis}"\
483       "${LFBFL_html_path_start}.4"
484
485     enhanced_pushd "${LFBFL_temp_path}" 2
486     can_continue_after_enhanced_pushd || return
487     sed -i\
488       -e '/@current_tree_light@/{r current_tree_light.txt' -e 'd}'\
489       -e '/@current_tree@/{r current_tree.txt' -e 'd}'\
490       "${LFBFL_repository_name}.html.4"
491     if [[ enhanced_pushd_result -eq 0 ]]; then
492       enhanced_popd "${LFBFL_temp_path}" 2\
493         || return
494     fi
495
496     insert_file_at_token "${LFBFL_html_path_start}.4"\
497       @files_listing_HTMLPreInput@\
498       "./${LFBFL_subdir2}/temp/files_listing.html.tpl"\
499       "${LFBFL_html_path_start}.5"
500
501     overwrite_if_not_equal "./${LFBFL_repository_name}.html"\
502       "${LFBFL_html_path_start}.5" 1
503     LFBFL_HTML_updated=$?
504   fi
```

```

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_HTML_updated -eq 0 ]]; then
    return
  fi

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"
    done
  else
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
        > /dev/null
    done
  fi

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_subdir2}/temp/"
  done
}

create_PDF "$@"
```

2.11 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    echo "$3"
    cat "$1"
    echo "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
  readonly LFBFL_temp
  if [[ -n "$5" ]]; then
```

```
    echo "$5"\
      > "${LFBFL_temp}"
  fi
  sed -e "s/^/$3/g" "$1"\
    >> "${LFBFL_temp}"
  if [[ -n "$4" ]]; then
    eval "$4"
  fi
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

split_file_in_two(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_part1
  # $4=$file_name_part2
  declare -ir LFBFL_line_number=$(
    grep -n "$2" "$1"\
    | head --lines=1\
    | cut -f 1 -d ':'
  )
  declare -ir LFBFL_line_count=$(ll_wc -l -n "$1")
  declare -ir LFBFL_lines_after=$((
    LFBFL_line_count - LFBFL_line_number
  ))
  head --lines="$((LFBFL_line_number - 1))" "$1"\
    > "$3"
  tail --lines="${LFBFL_lines_after}" "$1"\
    > "$4"
}

insert_file_at_token(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_to_insert
  # $4=$file_name_result
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
```

```
    "${LFBFL_end_file_name}"
  cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}"
  fi
}
```

2.12 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
```

```
    '**/*.htm'
    '**/*.html'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.tex'
    '*.toml'
    '*.tpl'
```

```
    '*.ts'
    '*.txt'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

2.13 build_and_checks_dependencies/grammar_and_spell_check.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
```

```
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spell_check(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_grammar_or_spell_checker_command=""
  grep_variable "$1" grammar_or_spell_checker_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_or_spell_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    return
  fi
  local LFBFL_file_path
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 0 ]]\
      || echo "Iterating on pattern: ${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | while read -r LFBFL_file_path;
    do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

2.14 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
```

```
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# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
```

```
49
50    enhanced_set_shell_option pipefail\
51      && trap 'enhanced_unset_shell_option pipefail' RETURN
52
53    # Source
54    local LFBFL_license_source_subdir
55    LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
56    readonly LFBFL_license_source_subdir
57    local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
58    LFBFL_source_license_prefix+="license_file_header_"
59    readonly LFBFL_source_license_prefix
60    # Target
61    local LFBFL_license_target_subdir
62    LFBFL_license_target_subdir="${LFBFL_work_directory}/"
63    LFBFL_license_target_subdir="build_and_checks_variables/temp/"
64    readonly LFBFL_license_target_subdir
65    local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
66    LFBFL_target_license_prefix+="license_file_header_"
67    readonly LFBFL_target_license_prefix
68
69    declare -ar LFBFL_licenses=(
70      "AGPLv3+"
71      "GPLv3+"
72      "LGPLv3+"
73    )
74
75    # ------------------------------------------------------------------
76    # Various language definitions for text file formats
77    declare -ar LFBFL_block_comment_languages=(
78      "c"
79      "py"
80    )
81    declare -ar LFBFL_block_comment_enters=(
82      '/*'
83      '"""'
84    )
85    declare -ar LFBFL_block_comment_exits=(
86      '*/'
87      '"""'
88    )
89    equal "${#LFBFL_block_comment_languages[@]}"\
90          "${#LFBFL_block_comment_enters[@]}"\
91          "${#LFBFL_block_comment_exits[@]}"
92    declare -i LFBFL_equal_result=$?
93    if [[ LFBFL_equal_result -eq 0 ]]; then
94      echo $'/!\\Problem with the definitions of languages arrays 1/!\\'
```

```
fi

declare -ar LFBFL_line_comment_languages=(
  "sh"
  "tex"
)
declare -ar LFBFL_line_comment_prefixes=(
  '# '
  '% '
)
equal "${#LFBFL_line_comment_languages[@]}"\
      "${#LFBFL_line_comment_prefixes[@]}"
LFBFL_equal_result=$?
if [[ LFBFL_equal_result -eq 0 ]]; then
  echo $'/!\\Problem with the definitions of languages arrays 2/!\\'
fi
# ------------------------------------------------------------------

# ------------------------------------------------------------------
# We generate the license header templates adapted to various text
# files formats.
# Kamoulox do endfor, Jacques Beauregard XD
# (C'est toujours mieux que l'inverse :) !)
# for do endfor -> for do done (Fort Doux Donne).
local LFBFL_some_license
local LFBFL_license_prefix
local LFBFL_license_prefix2
local LFBFL_extension
local LFBFL_license_file_name
local LFBFL_enter_string
local LFBFL_exit_string
local LFBFL_temp2
local LFBFL_prefix_string
local LFBFL_intermediate_file_name
local LFBFL_file_prefix
declare -i LFBFL_generate_from_template_result
for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
  LFBFL_license_prefix="${LFBFL_source_license_prefix}"
  LFBFL_license_prefix+="${LFBFL_some_license}"
  LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
  LFBFL_license_prefix2+="${LFBFL_some_license}"
  for ((i=0; i<${#LFBFL_block_comment_languages[@]}; i++)); do
    LFBFL_extension=${LFBFL_block_comment_languages[i]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}"
    LFBFL_license_file_name+=".${LFBFL_extension}"
    LFBFL_enter_string=${LFBFL_block_comment_enters[i]}
```

```
141         LFBFL_exit_string=${LFBFL_block_comment_exits[i]}
142         generate_from_template_with_block_comments\
143           "${LFBFL_license_prefix}.tpl"\
144           "${LFBFL_license_file_name}.tpl"\
145           "${LFBFL_enter_string}" "${LFBFL_exit_string}"
146         LFBFL_generate_from_template_result=$?
147         if [[ LFBFL_i_verbose -eq 1 ]]; then
148           if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
149             echo\
150               "License template ${LFBFL_license_file_name}.tpl updated"
151           elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
152             echo\
153               "License template ${LFBFL_license_file_name}.tpl created"
154           fi
155         fi
156       done
157       LFBFL_temp2=".generate_from_template_with_line_comments.temp"
158       for ((i=0; i<${#LFBFL_line_comment_languages[@]}; i++)); do
159         LFBFL_extension=${LFBFL_line_comment_languages[i]}
160         LFBFL_license_file_name="${LFBFL_license_prefix2}"
161         LFBFL_license_file_name+=".${LFBFL_extension}"
162         LFBFL_prefix_string=${LFBFL_line_comment_prefixes[i]}
163         LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
164         LFBFL_intermediate_file_name+="${LFBFL_temp2}"
165         LFBFL_file_prefix=""
166         if [[ "${LFBFL_extension}" == "sh" ]]; then
167           LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
168         fi
169         # The sed command is here to remove trailing spaces of previously
170         # empty lines after comment prefix is added to the line.
171         generate_from_template_with_line_comments\
172           "${LFBFL_license_prefix}.tpl"\
173           "${LFBFL_license_file_name}.tpl"\
174           "${LFBFL_prefix_string}"\
175           "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
176           "${LFBFL_file_prefix}"
177         LFBFL_generate_from_template_result=$?
178         if [[ LFBFL_i_verbose -eq 1 ]]; then
179           if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
180             echo\
181               "License template ${LFBFL_license_file_name}.tpl updated"
182           elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
183             echo\
184               "License template ${LFBFL_license_file_name}.tpl created"
185           fi
186         fi
```

```
187     done
188   done
189   # ------------------------------------------------------------------
190
191   # Now that base license templates generated languages license
192   # templates in working directory, we can go there.
193   pushd_to_work_directory\
194     && trap 'popd_from_work_directory' RETURN
195   can_continue_after_enhanced_pushd || return
196
197   local LFBFL_data_file_name="build_and_checks_variables/"
198   LFBFL_data_file_name+="repository_data.txt"
199   readonly LFBFL_data_file_name
200   local LFBFL_repository_name=""
201   grep_variable "${LFBFL_data_file_name}" repository_name\
202     --result-variable-prefix="LFBFL_"\
203     --replace-line-returns-by=""
204   local LFBFL_license=""
205   grep_variable "${LFBFL_data_file_name}" license\
206     --result-variable-prefix="LFBFL_"
207   local LFBFL_license2=""
208   grep_variable "${LFBFL_data_file_name}" license2\
209     --result-variable-prefix="LFBFL_"
210   local LFBFL_author_full_name=""
211   grep_variable "${LFBFL_data_file_name}" author_full_name\
212     --result-variable-prefix="LFBFL_"\
213     --replace-line-returns-by=" "
214
215   LFBFL_license_prefix="build_and_checks_variables/temp/"
216   LFBFL_license_prefix+="license_file_header_"
217   readonly LFBFL_license_prefix
218   LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
219   readonly LFBFL_license_prefix2
220   declare -r\
221     LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
222
223   # First year according to current state of git repository.
224   declare -ir LFBFL_first_year=$(
225     git log\
226     | grep 'Date:'\
227     | cut -f 8 -d ' '\
228     | tail -1
229   )
230   # Last year according to current state of git repository.
231   declare -ir LFBFL_last_year=$(
232     git log\
```

```
233       | grep 'Date:'\
234       | cut -f 8 -d ' '\
235       | head -1
236     )
237     local LFBFL_copyright_string
238     LFBFL_copyright_string="${LFBFL_first_year}-${LFBFL_last_year}"
239     LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
240     readonly LFBFL_copyright_string
241
242     prepare_filled_license_file_block(){
243       # $1=license_file_name
244       sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
245           -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
246           "$1.tpl"\
247         > "$1.temp"
248       overwrite_if_not_equal "$1" "$1.temp"
249       declare -r LFBFL_result=$?
250       if [[ LFBFL_i_verbose -eq 1 ]]; then
251         if [[ LFBFL_result -eq 1 ]]; then
252           echo "License template $1 updated"
253         elif [[ LFBFL_result -eq 2 ]]; then
254           echo "License template $1 created"
255         fi
256       fi
257       head --lines=-1 "$1"\
258         | tail --lines=+2\
259         > "$1.temp"
260       # Strange world: tail --lines=-1 was working
261       # to exclude first line.
262       # I tested it months ago, and saw with my eyes the start and end
263       # block comments removed with:
264       # head --lines=-1 "$1" | tail --lines=-1.
265       # Now, man page says to use --lines=+2 instead.
266       # Welcome in the world where AI change your environment and your
267       # documentation on the fly...
268     }
269
270     declare -Ar LFBFL_all_block_comment_languages=(
271       ["c"]="c"
272       ["php"]="c"
273       ["py"]="py"
274     )
275
276     local LFBFL_key
277     local LFBFL_dest
278     local LFBFL_license_file_name2
```

```
279   declare -i LFBFL_not_subfile
280   declare -i LFBFL_not_subfile2
281   for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
282     LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
283     LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
284     prepare_filled_license_file_block "${LFBFL_license_file_name}"
285     if [[ -n "${LFBFL_license2}" ]]; then
286       LFBFL_license_file_name2="${LFBFL_license_prefix3}"
287       LFBFL_license_file_name2+=".${LFBFL_dest}"
288       prepare_filled_license_file_block "${LFBFL_license_file_name2}"
289     fi
290     find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
291       | relevant_find\
292       | while read -r LFBFL_file_name;
293     do
294       is_subfile "${LFBFL_file_name}"\
295         "${LFBFL_license_file_name}.temp"
296       LFBFL_not_subfile=$?
297       LFBFL_not_subfile2=1
298       if [[ -n "${LFBFL_license2}" ]]; then
299         is_subfile "${LFBFL_file_name}"\
300           "${LFBFL_license_file_name2}.temp"
301         LFBFL_not_subfile2=$?
302       fi
303       if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
304       then
305         echo "File ${LFBFL_file_name} has no/wrong license header?"
306       fi
307     done
308     rm "${LFBFL_license_file_name}.temp"
309     if [[ -n "${LFBFL_license2}" ]]; then
310       rm "${LFBFL_license_file_name2}.temp"
311     fi
312   done
313
314   prepare_filled_license_file_line(){
315     # $1=LFBFL_license_file_name
316     sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
317         -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
318         "$1.tpl"\
319       > "$1.temp"
320     overwrite_if_not_equal "$1" "$1.temp"
321     declare -r LFBFL_result=$?
322     if [[ LFBFL_i_verbose -eq 1 ]]; then
323       if [[ LFBFL_result -eq 1 ]]; then
324         echo "License template $1 updated"
```

```
      elif [[ LFBFL_result -eq 2 ]]; then
        echo "License template $1 created"
      fi
    fi
  }

  declare -Ar LFBFL_all_line_comment_languages=(
    ["sh"]="sh"
    ["tex"]="tex"
  )

  for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_line "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_line "${LFBFL_license_file_name2}"
    fi
    find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
      | relevant_find\
      | while read -r LFBFL_file_name;
    do
      is_subfile "${LFBFL_file_name}" "${LFBFL_license_file_name}"
      LFBFL_not_subfile=$?
      LFBFL_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_name}" "${LFBFL_license_file_name2}"
        LFBFL_not_subfile2=$?
      fi
      if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
      then
        echo "File ${LFBFL_file_name} has no/wrong license header?"
      fi
    done
  done
}

build_licenses_templates "$@"
```

2.15 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
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```

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```

2.16 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

2.17 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

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```

2.18 build_and_checks_dependencies/lines_counts.libr.sh

```
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```

```
# along with DevOrSysAdminScripts.
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && echo "Iterating on pattern: ${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H -v 'a(?!a)a'
  done
}

all_self_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
```

```
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  # below and another lines, see SC2145 false positive
  all_code_lines "$@" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  all_self_code_lines "$@" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  all_self_code_lines "$@" \
    | not_empty_lines_after_file_name
}
```

```
112
113 code_lines_count_all(){
114   # Options:
115   #   --verbose
116   # shellcheck disable=SC2034
117   declare -i LFBFL_i_verbose=0
118   get_verbose_option "$@"
119
120   enhanced_set_shell_option pipefail\
121     && trap 'enhanced_unset_shell_option pipefail' RETURN
122
123   declare -gi code_lines_count_all_result
124   code_lines_count_all_result=$(
125     all_self_code_lines\
126     | wc -l
127   )
128   # The echo is for command line use.
129   # A script should use the result value instead.
130   # shellcheck disable=SC2248
131   echo ${code_lines_count_all_result}
132 }
133
134 code_lines_count_empty(){
135   # Options:
136   #   --verbose
137   # shellcheck disable=SC2034
138   declare -i LFBFL_i_verbose=0
139   get_verbose_option "$@"
140
141   enhanced_set_shell_option pipefail\
142     && trap 'enhanced_unset_shell_option pipefail' RETURN
143
144   declare -gi code_lines_count_empty_result
145   # shellcheck disable=SC2119
146   code_lines_count_empty_result=$(
147     all_self_empty_code_lines\
148     | wc -l
149   )
150   # The echo is for command line use.
151   # A script should use the result value instead.
152   # shellcheck disable=SC2248
153   echo ${code_lines_count_empty_result}
154 }
155
156 code_lines_count_not_empty(){
157   # Options:
```

```
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc -l
  )
  # The echo is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  echo ${code_lines_count_not_empty_result}
}
```

2.19 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "lines_filters.sh" to
```

```
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # echo "$ll_wc_var_arg"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_number_only=1
    elif [[ ll_wc_var_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
```

```
  # typeset -p ll_wc_var_args
  # echo "$ll_wc_var_number_only"
  if [[ ll_wc_var_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # echo "normal"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# echo "https://domain/list-path/2024-05/msg00013.html" | \
#   sed -e 's|domain|lists.gnu.org|' \
#       -e 's|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

```

```
117 in_place_grep(){
118   declare -r LFBFL_temp="${!#}.in_place_grep.temp"
119   grep "$@" \
120     > "${LFBFL_temp}"
121   declare -ir LFBFL_lines_before=$(ll_wc -l -n "${!#}")
122   declare -ir LFBFL_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
123   # echo "$LFBFL_lines_before"
124   # echo "$LFBFL_lines_after"
125   if [[ "${LFBFL_lines_before}" ==  "${LFBFL_lines_after}" ]]; then
126     rm "${LFBFL_temp}"
127     return
128   fi
129   mv "${LFBFL_temp}" "${!#}"
130 }
131
132 grep_variable(){
133   # $1=$file
134   # $2=$variable_name
135   # Options :
136   #   --replace-line-returns-by=""
137   #   --result-variable-prefix="LFBFL_" for example
138   declare -r LFBFL_regexp="(?<=^$2=).*$"
139   local LFBFL_arg
140   # echo $LFBFL_regexp
141   local LFBFL_variable_value
142   if [[ "$*" == *--replace-line-returns-by=* ]]; then
143     local LFBFL_s_replace=""
144     for LFBFL_arg in "$@"; do
145       if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
146         LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
147         break
148       fi
149     done
150     LFBFL_variable_value=$(
151       grep -oP "${LFBFL_regexp}" "$1"\
152       | sed -Ez -e 's/\n/'"${LFBFL_s_replace}"'/Mg'
153     )
154     # We remove the trailing replace string,
155     # since the intent is only to replace the intermediate \n.
156     if [[ -n "${LFBFL_s_replace}" ]]; then
157     LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
158     fi
159   else
160     LFBFL_variable_value=$(grep -oP "${LFBFL_regexp}" "$1")
161   fi
162   local LFBFL_prefix=""
```

```
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
      LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
      break
    fi
  done
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # echo "${LFBFL_variable_name} : ${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # echo "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep -v '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep -v '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep -E '^( |$)'
}

not_spampty_lines(){
  grep '^[^ ]'
  # grep -v -E '^( |$)'
}

all_space_lines(){
  grep '^[ ]\+$'
}

```

```
not_all_space_lines(){
  grep -v '^[ ]\+$'
}

# All space or empty lines = aspampty
aspampty_lines(){
  grep '^[ ]*$'
}

not_aspampty_lines(){
  grep '[^ ]'
  # grep -v '^[ ]*$'
}

empty_lines_after_file_name(){
  grep '^[^:]\+:$'
}

not_empty_lines_after_file_name(){
  grep -v '^[^:]\+:$'
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep -v '^[^:]\+: '
}

spampty_lines_after_file_name(){
  grep -E '^[^:]+:( |$)'
}

not_spampty_lines_after_file_name(){
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep -v -E '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
```

```
}

not_all_space_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# :). Nice command :).

not_JS_dependencies_find(){
  grep -vE -e "(^|/)node_modules/"\
           -e "(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep -vE -e "^([^:]+/)?node_modules/"\
           -e "^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
  not_JS_dependencies_grep
}

not_python_cache_find(){
  grep -vE -e "(^|/)__pycache__/"\
           -e "(^|/)\.mypy_cache/"\
           -e "(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
  grep -vE -e "^([^:]+/)?__pycache__/"\
```

```
            -e "^([^:]+/)?\.mypy_cache/"\
            -e "^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_python_cache_find
}

not_cache_grep(){
  not_python_cache_grep
}

not_git_find(){
  grep -vE "(^|/)\.git/"
}

not_git_grep(){
  grep -vE "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep -vE "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep -vE "^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep -vE -e "(^|/)COPYING$"\
           -e "(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep -vE -e "^([^:]+/)?COPYING:"\
           -e "^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.tex$"
}
```

```
not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.html$"
}

not_main_html_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.html:"
}

not_temp_file_find(){
  grep -vE "(^|/)build_and_checks_variables/temp/.*$"
}

not_temp_file_grep(){
  grep -vE "^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
}

relevant_find(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_find\
    | not_cache_find\
    | not_git_find\
    | not_archive_find\
    | not_temp_file_find
}
```

```
relevant_grep(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep
}
```

2.20 build_and_checks_dependencies/lines_maps.libr.sh

```
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
```

```
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
      -e 's/\xe2\x80\x80/ /g'\
      -e 's/\xe2\x80\x81/ /g'\
      -e 's/\xe2\x80\x82/ /g'\
      -e 's/\xe2\x80\x83/ /g'\
      -e 's/\xe2\x80\x84/ /g'\
      -e 's/\xe2\x80\x85/ /g'\
      -e 's/\xe2\x80\x86/ /g'\
      -e 's/\xe2\x80\x87/ /g'\
      -e 's/\xe2\x80\x88/ /g'\
      -e 's/\xe2\x80\x89/ /g'\
      -e 's/\xe2\x80\x8a/ /g'\
      -e 's/\xe2\x80\x8b/ /g'\
      -e 's/\xe2\x80\x8c/ /g'\
      -e 's/\xe2\x80\x8d/ /g'\
      -e 's/\xe2\x80\x8e/ /g'\
      -e 's/\xe2\x80\x8f/ /g'\
      -e 's/\xe2\x80\xa8/ /g'\
      -e 's/\xe2\x80\xa9/ /g'\
      -e 's/\xe2\x80\xaa/ /g'\
      -e 's/\xe2\x80\xab/ /g'\
      -e 's/\xe2\x80\xac/ /g'\
      -e 's/\xe2\x80\xad/ /g'\
      -e 's/\xe2\x80\xae/ /g'\
      -e 's/\xe2\x80\xaf/ /g'\
      -e 's/\xe2\x81\x9f/ /g'\
      -e 's/\xe2\x81\xa0/ /g'\
      -e 's/\xe2\x81\xa1/ /g'\
      -e 's/\xe2\x81\xa2/ /g'\
      -e 's/\xe2\x81\xa3/ /g'\
      -e 's/\xe2\x81\xa4/ /g'\
      -e 's/\xe2\x81\xa5/ /g'\
      -e 's/\xe2\x81\xa6/ /g'\
      -e 's/\xe2\x81\xa7/ /g'\
      -e 's/\xe2\x81\xa8/ /g'\
      -e 's/\xe2\x81\xa9/ /g'\
      -e 's/\xe2\x81\xaa/ /g'\
      -e 's/\xe2\x81\xab/ /g'\
      -e 's/\xe2\x81\xac/ /g'\
      -e 's/\xe2\x81\xad/ /g'\
      -e 's/\xe2\x81\xae/ /g'\
      -e 's/\xe2\x81\xaf/ /g'
}
```

2.21 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```bash
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=$target_file_name
  # $2=$temp_file_name
  # $3=$keep_temp
  # $4=$tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories.
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  if [[ ! -f "$1" ]]; then
    if [[ -n "$3" ]]; then
      cp -p "$2" "$1"
    else
      mv "$2" "$1"
    fi
    return 2
  fi
  declare -i LFBFL_is_equal
```

```
  if [[ -n "$4" ]]; then
    # Not setting pipefail since the result would still be incorrect.
    # shellcheck disable=SC2312
    diff "$1" "$2"\
      | grep -E "^(>|<)"\
      | grep -v "/$"
    LFBFL_is_equal=${PIPESTATUS[2]}
  else
    diff -q "$1" "$2"
    LFBFL_is_equal=1-$?
  fi
  if [[ LFBFL_is_equal -eq 1 ]]; then
    if [[ -z "$3" ]]; then
      rm "$2"
    fi
    return 0
  fi
  if [[ -n "$3" ]]; then
    cp -p "$2" "$1"
  else
    mv "$2" "$1"
  fi
  return 1
}
```

2.22 build_and_checks_dependencies/python_black_complement.libr.sh

```
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```

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  enhanced_set_bash_option globstar\
    && trap 'enhanced_unset_bash_option globstar' RETURN

  [[ LFBFL_i_verbose -eq 1 ]]\
    && echo "Checking empty lines after Python function docstrings"
  pcre2grep -M --\
    $'def ([^"]|"(?!""))*"""([^"]|"(?!""))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
```

```
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  check_no_empty_line_after_python_function_docstring "$@"
}
```

2.23 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
```

```
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  [[ LFBFL_i_verbose -eq 1 ]]\
    && echo "Checking import of _collections_abc is at the right place"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_temp=".check_collections_abc_place.temp"
  local LFBFL_file_name
  find . -name "*.py"\
    | relevant_find\
    | while read -r LFBFL_file_name;
  do
    [[ -f "${LFBFL_file_name}" ]] || continue
    sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
      "${LFBFL_file_name}"\
      > "${LFBFL_file_name}${LFBFL_temp}"
    overwrite_if_not_equal "${LFBFL_file_name}"\
      "${LFBFL_file_name}${LFBFL_temp}"
  done
}

python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  check_collections_abc_place "$@"
}
```

2.24 build_and_checks_dependencies/repository_data.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_name
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_bash_option extglob\
    && trap 'enhanced_unset_bash_option extglob' RETURN

  grep_variable "$1"\
```

```
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && echo "You say \"${LFBFL_upgrade_venvs_time_interval_in_seconds}\"."

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) echo "No wat for you?";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && echo "I say \"${LFBFL_upgrade_venvs_time_interval_in_seconds}\"."
}
```

2.25 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php
/*
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
```

```
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".
*/

require_once('./build_and_checks_dependencies/split_score.libr.php');

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// The first cast "(int) generate_split_score" is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1], (int) $argv[2], explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
```

2.26 build_and_checks_dependencies/split_score.libr.php

```
<?php
/*
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"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".
*/

function generate_split_score(
  $b_after_before,
  $i_max_length,
  $a_delimiter_strings_domain,
){
  return function(
    $s_delimiter_string,
    $i_cut_position,
    $b_is_cut_after,
  ) use ($b_after_before, $i_max_length, $a_delimiter_strings_domain){
    if(
      !in_array(
        $s_delimiter_string, $a_delimiter_strings_domain, true
      )
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_after_before){
      // When $b_after_before,
      // always way better to cut after '/' than before it.
      // When !$b_after_before (aka before_after),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
```

```
}
```

2.27 build_and_checks_dependencies/strings_functions.libr.sh

```
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# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"

split_line_at(){
  # $1=$line
  # $2=$position
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # echo "$split_line_at_result_beginning"
```

```
43    # echo "$split_line_at_result_end"
44  }
45
46  declare -gir SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM=1
47  declare -gir SPLIT_SCORE_PROPERTY_POSITION_UNIFORM=2
48  declare -gir SPLIT_SCORE_PROPERTY_LARGER_AFTER=4
49  declare -gir SPLIT_SCORE_PROPERTY_LARGER_BEFORE=8
50
51  split_score_simple(){
52    # $1=$after_before
53    # $2=$delimiters_strings_domain concatenated characters/delimiters
54    # $3=$delimiter_string a single character
55    # $4=$is_cut_after
56    declare -gi split_score_result
57    if [[ "$2" != *$3* ]]; then
58      split_score_result=0
59      return
60    fi
61    if [[ "$4" == "$1" ]]; then
62      split_score_result=2
63      return
64    fi
65    split_score_result=1
66  }
67
68  get_split_score_simple(){
69    # $1=$after_before
70    # $2=$max_length
71    # $3=$delimiters_strings_domain
72    declare -g get_split_score_result="split_score_simple $1 '$3'"
73    declare -gi get_split_score_result2
74    # get_split_score_result2=$((11-$1*4))
75    # if [[ $1 -eq 1 ]]; then
76    #   get_split_score_result2=7
77    # else
78    #   get_split_score_result2=11
79    # fi
80    get_split_score_result2=$((
81      SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM
82      + SPLIT_SCORE_PROPERTY_POSITION_UNIFORM
83      + SPLIT_SCORE_PROPERTY_LARGER_BEFORE
84      + (
85        $1 * (
86          SPLIT_SCORE_PROPERTY_LARGER_AFTER
87          - SPLIT_SCORE_PROPERTY_LARGER_BEFORE
88        )
```

```
    )
  ))
}

get_split_score_exec(){
  declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
  SPLIT_SCORE_EXEC+="/split_score.exec.php"
}

split_score(){
  # $1=$after_before
  # $2=$max_length
  # $3=$delimiters_strings_domain concatenated characters/delimiters
  # $4=$delimiter_string a single character
  # $5=$cut_position
  # $6=$is_cut_after
  get_split_score_exec
  declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
  declare -gi split_score_result
  split_score_result=$(eval "${LFBFL_command}")
}

# This function is currently not called,
# hence we need that shellcheck disable.
# shellcheck disable=SC2034
get_split_score(){
  # $1=$after_before
  # $1=$max_length
  # $2=$delimiters_strings_domain
  declare -g get_split_score_result="split_score $1 $2 '$3'"
  declare -gi get_split_score_result2
  # get_split_score_result2=$((9-$1*4))
  get_split_score_result2=$((
    SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM
    + SPLIT_SCORE_PROPERTY_LARGER_BEFORE
    + (
      $1 * (
        SPLIT_SCORE_PROPERTY_LARGER_AFTER
        - SPLIT_SCORE_PROPERTY_LARGER_BEFORE
      )
    )
  ))
}

split_line_at_most(){
  # $1=$line
```

```
135     # $2=$max_length of beginning result string
136     # $3=$split_score_command
137     #   negative score means do not consider as a possible split
138     # $4=$split_score_command_properties :
139     #   These properties are flags.
140     #   If someday some property is not a flag,
141     #   then create split_score_command_properties2.
142     #   - null or 0 no property;
143     #   - 1 split score doesn't depend on which delimiter matches;
144     #   - 2 split score doesn't depend on the current position;
145     #   - 4 any split score after is always larger
146     #     than any split score before;
147     #   - 8 any split score before is always larger
148     #     than any split score after;
149     # $5=$forbidden_previous_character
150     # $6=$split_score_command_delimiter_strings_domain
151     #   optimisation if small domain
152     #   - this argument can be "null", then we will use $3 only instead,
153     #     in that case, only mono-character strings are considered,
154     #   - this argument can be an array of characters,
155     #     optimize accordingly,
156     #   - this argument can be an array of strings,
157     #     optimize accordingly,
158     #   - this argument can be an array of regexps,
159     #     optimize accordingly.
160     #     Not sure an array of regexps makes more sense than a single
161     #     one. Maybe they could be both cases with corresponding
162     #     optimizations. Same for a single character or substring.
163     #
164     # Note that we're always looking for the last position to split
165     # among maximum score;
166     # if some position gets as score from 2 delimiters,
167     # its score is always the maximum of the two corresponding scores.
168     declare -g split_line_at_most_result_start
169     declare -g split_line_at_most_result_end
170     declare -A LFBFL_positions
171     # echo "$1 $2 $3 $4"
172     # For my use case in bash scripts, I will need only an array of
173     # characters. See get_split_score().
174     if [[ -n "$6" ]]; then
175       echo "split_line_at_most() \$4 NOT IMPLEMENTED YET"
176     fi
177     if [[ $2 -ge ${#1} ]]; then
178       split_line_at_most_result_start=$1
179       split_line_at_most_result_end=""
180       return
```

```
181     fi
182     declare -r LFBFL_sort_command="sort --numeric-sort"
183     declare -ir LFBFL_length_minus_1=$((${#1} - 1))
184     declare -ir LFBFL_i_max=$(
185       min "${LFBFL_sort_command}" "${LFBFL_length_minus_1}" "$2"
186     )
187     declare -i LFBFL_i
188     declare -i LFBFL_j
189     local LFBFL_current_char
190
191     # At the beginning, each position has score 0.
192     for ((LFBFL_i=0; LFBFL_i<=LFBFL_i_max; ++LFBFL_i)) do
193       LFBFL_positions["${LFBFL_i}"]="0"
194     done
195     LFBFL_j=$((LFBFL_i+1))
196     LFBFL_positions["${LFBFL_j}"]="0"
197
198     # If we have a forbidden previous character,
199     # we mark the forbidden positions with -1.
200     if [[ -n "$5" ]]; then
201       for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
202         LFBFL_j=$((LFBFL_i+1))
203         LFBFL_current_char="${1:${LFBFL_i}:1}"
204         if [[ "${LFBFL_current_char}" == "$5" ]]; then
205           LFBFL_positions["${LFBFL_j}"]="-1"
206         fi
207       done
208     fi
209
210     if [[ "$4" == "7" ]]; then
211       for ((LFBFL_j=LFBFL_i_max; LFBFL_j>0; --LFBFL_j)) do
212         LFBFL_i=$((LFBFL_j-1))
213         LFBFL_current_char="${1:${LFBFL_i}:1}"
214         eval "$3 '${LFBFL_current_char}' 0"
215         # echo "${split_score_result}|${LFBFL_i}"
216         if [[ split_score_result -ge 1 ]]; then
217           # echo "${split_score_result}|${LFBFL_i}"
218           if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
219             LFBFL_positions["${LFBFL_i}"]=$(
220               max "${LFBFL_sort_command}"\
221                 "${LFBFL_positions["${LFBFL_i}"]}"\
222                 "${split_score_result}"
223             )
224           fi
225         fi
226         eval "$3 '${LFBFL_current_char}' 1"
```

```
227       # echo "${split_score_result}|${LFBFL_j}"
228       if [[ split_score_result -ge 1 ]]; then
229         # echo "${split_score_result}|${LFBFL_j}"
230         if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
231           LFBFL_positions["${LFBFL_j}"]=$(
232             max "${LFBFL_sort_command}"\
233               "${LFBFL_positions["${LFBFL_j}"]}"\
234               "${split_score_result}"
235           )
236           break
237         fi
238       fi
239     done
240   else
241     for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
242       LFBFL_j=$((LFBFL_i+1))
243       LFBFL_current_char="${1:${LFBFL_i}:1}"
244       eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
245       # echo "${split_score_result}|${LFBFL_i}"
246       if [[ split_score_result -ge 1 ]]; then
247         # echo "${split_score_result}|${LFBFL_i}"
248         if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
249           LFBFL_positions["${LFBFL_i}"]=$(
250             max "${LFBFL_sort_command}"\
251               "${LFBFL_positions["${LFBFL_i}"]}"\
252               "${split_score_result}"
253           )
254         fi
255       fi
256       eval "$3 '${LFBFL_current_char}' ${LFBFL_j} 1"
257       # echo "${split_score_result}|${LFBFL_j}"
258       if [[ split_score_result -ge 1 ]]; then
259         # echo "${split_score_result}|${LFBFL_j}"
260         if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
261           LFBFL_positions["${LFBFL_j}"]=$(
262             max "${LFBFL_sort_command}"\
263               "${LFBFL_positions["${LFBFL_j}"]}"\
264               "${split_score_result}"
265           )
266         fi
267       fi
268     done
269   fi
270   declare -ir LFBFL_max_score=$(
271     max 'sort --numeric-sort' "${LFBFL_positions[@]}"
272   )
```

```
  local LFBFL_value
  declare -i LFBFL_best_position
  for ((LFBFL_i=0; LFBFL_i<=LFBFL_i_max; ++LFBFL_i)) do
    local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
    # echo "${LFBFL_i} ${LFBFL_value}"
    if [[ "${LFBFL_value}" == "${LFBFL_max_score}" ]]; then
      LFBFL_best_position=$((LFBFL_i))
    fi
  done
  # shellcheck disable=SC2248
  split_line_at "$1" ${LFBFL_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # echo "$split_line_at_most_result_start"
  # echo "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=$new_lines
  # $2=$prefix
  # $3=$max_length
  # $4=$suffix
  # $5=$split_score_command
  # $6=$split_score_command_properties
  # $7=$forbidden_previous_character ('\' usually)
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
```

```
319   enhanced_set_shell_option pipefail\
320     && trap 'enhanced_unset_shell_option pipefail' RETURN
321
322   declare -g split_last_line_result="$1"
323   declare -ir LFBFL_max_length_plus=$(($3 + 1))
324   declare -ir LFBFL_length2=$(($3 - ${#4}))
325   declare -r LFBFL_regexp='.\{'"${LFBFL_max_length_plus}"'\}$'
326   if
327     echo "$1"\
328       | sed -e 's/\\n/\n/g'\
329       | grep -q "${LFBFL_regexp}";
330   then
331     declare -r LFBFL_start=$(
332       echo "$1"\
333       | sed -e 's/\\n/\n/g'\
334       | head --lines=-1\
335       | sed -z 's/\n/\\n/g'
336     )
337     # echo "start: $LFBFL_start"
338     declare -r LFBFL_last_line=$(
339       echo "$1"\
340       | sed -e 's/\\n/\n/g'\
341       | tail --lines=1
342     )
343     # echo "last_line: ${LFBFL_last_line}"
344     split_last_line_result=""
345     if [[ -n "${LFBFL_start}" ]]; then
346       split_last_line_result="${LFBFL_start}\n"
347     fi
348     if [[ -n "$5" ]]; then
349       split_line_at_most "${LFBFL_last_line}" "${LFBFL_length2}"\
350         "$5" "$6" "$7"
351       split_last_line_result+="${split_line_at_most_result_start}$4"
352       split_last_line_result+="\n"
353       split_last_line_result+="$2${split_line_at_most_result_end}"
354     else
355       split_last_line_result+="${LFBFL_last_line:0:${LFBFL_length2}}"
356       split_last_line_result+="$4\n"
357       split_last_line_result+="$2${LFBFL_last_line:${LFBFL_length2}}"
358     fi
359   fi
360 }
361
362 repeated_split_last_line(){
363   declare -g repeated_split_last_line_result="$1"
364   local LFBFL_result="$1x"
```

```
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # echo "${repeated_split_last_line_result}"
  done
}
```

2.28 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
```

```
34
35 too_long_code_lines(){
36   # Options:
37   #   --verbose
38   #   --work-directory=""
39   #   --max-line-length
40   declare -i LFBFL_i_verbose=0
41   get_verbose_option "$@"
42   local LFBFL_work_directory=""
43   get_work_directory_option "$@"
44   pushd_to_work_directory\
45     && trap 'popd_from_work_directory' RETURN
46   can_continue_after_enhanced_pushd || return
47
48   local LFBFL_arg
49   declare -i LFBFL_max_line_length=70
50   for LFBFL_arg in "$@"; do
51     if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
52       LFBFL_arg=${LFBFL_arg#--max-line-length=}
53       LFBFL_max_line_length=$((LFBFL_arg))
54       break
55     fi
56   done
57
58   enhanced_set_shell_option pipefail\
59     && trap 'enhanced_unset_shell_option pipefail' RETURN
60
61   get_COMMON_TEXT_FILES_GLOB_PATTERNS
62   local LFBFL_pattern
63   local LFBFL_long_line
64   local LFBFL_file_name
65   local LFBFL_line
66   local LFBFL_extension
67   local LFBFL_base_name
68   declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
69   for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
70     [[ LFBFL_i_verbose -eq 0 ]]\
71       || echo "Iterating on pattern: ${LFBFL_pattern}"
72     # shellcheck disable=SC2248
73     find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
74       | relevant_find\
75       | not_license_find\
76       | xargs grep -H '.\{'${LFBFL_overlength}'\}'\
77       | while read -r LFBFL_long_line;
78     do
79       LFBFL_file_name=${LFBFL_long_line%%:*} # Drop long ':*' suffix
```

```
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_name##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_name%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.tpl"
        then
          continue
        fi
        echo "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if ! [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
          echo "${LFBFL_long_line}"
        fi
      else
        echo "${LFBFL_long_line}"
      fi
    done
  done
}
```

2.29 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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```
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# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose=1
  fi
  readonly LFBFL_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_verbose -eq 1 ]] && echo "pipefail option activated"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp "./${LFBFL_common_file_name}.tpl"\
    "./${LFBFL_common_file_name}.temp"

  declare -ar LFBFL_file_names=(
    "build_md_from_printable_md.exec.sh"
    "check_shell_scripts_beginnings.libr.sh"
    "check_shell_scripts_indentation.libr.sh"
    "check_URLs.libr.sh"
    "common_options.libr.sh"
    "comparisons.libr.sh"
    "create_PDF.exec.sh"
    "generate_from_template.libr.sh"
    "get_common_text_glob_patterns.libr.sh"
    "grammar_and_spell_check.libr.sh"
    "licenses_templates/build_licenses_templates.exec.sh"
    "licenses_templates/license_file_header_AGPLv3+.tpl"
    "licenses_templates/license_file_header_GPLv3+.tpl"
    "licenses_templates/license_file_header_LGPLv3+.tpl"
    "lines_counts.libr.sh"
    "lines_filters.libr.sh"
    "lines_maps.libr.sh"
    "update_or_check_files_names_listing.exec.sh"
```

```
    "overwrite_if_not_equal.libr.sh"
    "python_black_complement.libr.sh"
    "python_isort_complement.libr.sh"
    "repository_data.libr.sh"
    "split_score.exec.php"
    "split_score.libr.php"
    "strings_functions.libr.sh"
    "too_long_code_lines.libr.sh"
  )

  local LFBFL_file_name
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_file_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
      | cut -f1 -d' '
    )
    LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    sed -i\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut -f1 -d' '
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
```

```
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

2.30 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
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# or (at your option) any later version.
#
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# that it will be useful,
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# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
```

```
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return

  declare -i LFBFL_write=0
  if [[ "$*" == *--write* ]]; then
    LFBFL_write=1
  fi
  readonly LFBFL_write

  declare -i LFBFL_append=0
  if [[ "$*" == *--append* ]]; then
    LFBFL_append=1
  fi
  readonly LFBFL_append

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
  readonly LFBFL_listing
  local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
  readonly LFBFL_data_file_name

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"
```

```
78
79    if [[ LFBFL_write -eq 1 ]]; then
80      : > "${LFBFL_listing}"
81    fi
82    # Remove line returns that are here to keep lines short.
83    sed -Ez 's/\\\n//Mg' "${LFBFL_listing}"\
84      > "${LFBFL_listing}.temp"
85    # shopt -s dotglob was needed at some point but I don't see why now.
86    # shellcheck disable=SC2248
87    get_split_score_simple 1 ${LFBFL_max_line_length} /
88    declare -r LFBFL_split_score_command="${get_split_score_result}"
89    local LFBFL_split_score_command_properties
90    LFBFL_split_score_command_properties="${get_split_score_result2}"
91    readonly LFBFL_split_score_command_properties
92    local LFBFL_suffix=\\\\ # instead of '\\' to avoid shellcheck SC1003
93    readonly LFBFL_suffix
94    local LFBFL_file_name
95    local LFBFL_base_file_name
96    local LFBFL_split_file_name
97    find . -type f -printf '%P\n'\
98      | relevant_find\
99      | sort\
100     | while read -r LFBFL_file_name;
101   do
102     git check-ignore -q "${LFBFL_file_name}" && continue
103     LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
104     [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
105       || continue
106     if [[ "${LFBFL_base_file_name}" == *.md ]]; then
107       if [[ -f "${LFBFL_file_name}.tpl" ]]; then
108         continue
109       fi
110     fi
111     # shellcheck disable=SC2248
112     repeated_split_last_line "${LFBFL_file_name}"\
113       ""\
114       ${LFBFL_max_line_length}\
115       "${LFBFL_suffix}"\
116       "${LFBFL_split_score_command}"\
117       "${LFBFL_split_score_command_properties}"
118     LFBFL_split_file_name="${repeated_split_last_line_result}"
119     if [[ LFBFL_write -eq 1 ]]; then
120       # shellcheck disable=SC2001
121       echo "${LFBFL_file_name}"\
122         | sed -e "s|${LFBFL_file_name}|${LFBFL_split_file_name}|g"\
123         >> "${LFBFL_listing}"
```

```
      continue
    fi
    if grep -q "${LFBFL_file_name}\$" "${LFBFL_listing}.temp"; then
      :
    else
      echo\
      "The file ${LFBFL_file_name} is not listed in ${LFBFL_listing}."
      if [[ LFBFL_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        echo "${LFBFL_file_name}"\
          | sed -e "s|${LFBFL_file_name}|${LFBFL_split_file_name}|g"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  while read -r LFBFL_file_name;
  do
    [[ "${LFBFL_file_name}" != '//'* ]] || continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    if ! [[ -f "${LFBFL_file_name}" ]]; then
      echo\
      "The non-file ${LFBFL_file_name} is listed in ${LFBFL_listing}."
    fi
  done < "${LFBFL_listing}.temp"

  rm "${LFBFL_listing}.temp"

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    cat "${LFBFL_listing}"
  fi
}

update_or_check_files_names_listing "$@"
```

2.31 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts

```

2.32 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

2.33 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

2.34 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

2.35 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

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you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

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but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

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-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
```

```
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
```

```
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
  content: counter(level1) "." counter(level2) " ";
  counter-increment: level2;
}
</style>
<title>@repository_name@</title>
</head>
<body >
<table class="title">
  <tr>
    <td style="padding:1ex">
      <h1 class="titlemain">@repository_name@</h1>
      <h2 class="titlerest">
        @author_full_name@
        <br>
        <a href="@author_website@">
          <span style="font-family:monospace">
          @author_website@
          </span>
        </a>
        <br>
        <a href="mailto:@author_email@">
          <span style="font-family:monospace">
          @author_email@
          </span>
        </a>
      </h2>
    </td>
  </tr>
</table>

<blockquote class="abstract">
  <span style="font-weight:bold">Abstract:</span>
  @abstract@
</blockquote>
<p>Current version: @current_date@</p>
<p>Current number of commits: @number_of_commits@</p>
<p>Current git SHA1: @current_git_SHA1@</p>
<p>
Code lines: @number_of_lines@
</p>


```

```
<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Listing of files</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
```

```
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

<p>
Bash can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/bash/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/bash/</span></a>,<br>
<a href="https://tiswww.case.edu/php/chet/bash/bashtop.html"><span
  style="font-family:monospace"
>https://tiswww.case.edu/php/chet/bash/bashtop.html</span></a>,<br>
<a href="https://savannah.gnu.org/projects/bash/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/bash/</span></a>.
</p>

```

```
<p>
GNU coreutils can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/coreutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/coreutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/coreutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/coreutils/</span></a>.
</p>

<p>
uutils coreutils can be found at the following URLs:<br>
<a href="https://uutils.github.io/coreutils/"><span
  style="font-family:monospace"
>https://uutils.github.io/coreutils/</span></a>,<br>
<a href="https://github.com/uutils/coreutils"><span
  style="font-family:monospace"
>https://github.com/uutils/coreutils</span></a>.
</p>

<p>
diff (GNU diffutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/diffutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/diffutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/diffutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/diffutils/</span></a>.
</p>

<p>
find, xargs (GNU findutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/findutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/findutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/findutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/findutils/</span></a>.
</p>

<p>
grep (GNU grep) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/grep/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/grep/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/grep/"><span
```

```
  style="font-family:monospace"
>https://savannah.gnu.org/projects/grep/</span></a>.
</p>

<p>
sed (GNU sed) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/sed/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/sed/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/sed/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/sed/</span></a>.
</p>

<p>
ShellCheck can be found at the following URLs:<br>
<a href="https://www.shellcheck.net/"><span
  style="font-family:monospace"
>https://www.shellcheck.net/</span></a>,<br>
<a href="https://github.com/koalaman/shellcheck"><span
  style="font-family:monospace"
>https://github.com/koalaman/shellcheck</span></a>.
</p>

<p>
GNU Wget can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/wget/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/wget/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/wget/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/wget/</span></a>.
</p>

<p>
more of util-linux can be found at the following URLs:<br>
<a href="https://www.kernel.org/pub/linux/utils/util-linux/"><span
  style="font-family:monospace"
>https://www.kernel.org/pub/linux/utils/util-linux/</span></a>,<br>
<a
href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
><span
  style="font-family:monospace"
>https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
></a>,<br>
<a href="https://github.com/util-linux/util-linux"><span
```

```
  style="font-family:monospace"
>https://github.com/util-linux/util-linux</span></a> official?
</p>

<p>
pandoc can be found at the following URLs:<br>
<a href="https://pandoc.org/"><span
  style="font-family:monospace"
>https://pandoc.org/</span></a>,<br>
<a href="https://github.com/jgm/pandoc"><span
  style="font-family:monospace"
>https://github.com/jgm/pandoc</span></a>.
</p>

<p>
pdftex/pdflatex can be found at the following URLs:<br>
<a href="https://www.tug.org/applications/pdftex/"><span
  style="font-family:monospace"
>https://www.tug.org/applications/pdftex/</span></a>,<br>
<a href="https://tug.org/texlive/svn/"><span
  style="font-family:monospace"
>https://tug.org/texlive/svn/</span></a>.
</p>

<p>
tree can be found at the following URLs:<br>
<a href="https://oldmanprogrammer.net/source.php?dir=projects/tree"
><span style="font-family:monospace"
>https://oldmanprogrammer.net/source.php?dir=projects/tree</span></a>,<br>
<a href="https://gitlab.com/OldManProgrammer/unix-tree"><span
  style="font-family:monospace"
>https://gitlab.com/OldManProgrammer/unix-tree</span></a>,<br>
<a href="https://github.com/Old-Man-Programmer/tree"><span
  style="font-family:monospace"
>https://github.com/Old-Man-Programmer/tree</span></a>.
</p>

<p>
git can be found at the following URLs:<br>
<a href="https://git-scm.com/about"><span
  style="font-family:monospace"
>https://git-scm.com/about</span></a>,<br>
<a href="https://github.com/git/git"><span
  style="font-family:monospace"
>https://github.com/git/git</span></a>.
</p>
```

```
<p>
pylint can be found at the following URLs:<br>
<a href="https://www.pylint.org/"><span
  style="font-family:monospace"
>https://www.pylint.org/</span></a>,<br>
<a href="https://pylint.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://pylint.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/pylint-dev/pylint"><span
  style="font-family:monospace"
>https://github.com/pylint-dev/pylint</span></a>,<br>
<a href="https://pypi.org/project/pylint/"><span
  style="font-family:monospace"
>https://pypi.org/project/pylint/</span></a>.
</p>

<p>
notify-send can be found at the following URLs:<br>
<a href="https://gitlab.gnome.org/GNOME/libnotify/"><span
  style="font-family:monospace"
>https://gitlab.gnome.org/GNOME/libnotify/</span></a>,<br>
<a href="https://gnome.pages.gitlab.gnome.org/libnotify/"><span
  style="font-family:monospace"
>https://gnome.pages.gitlab.gnome.org/libnotify/</span></a>.
</p>

<p>
black can be found at the following URLs:<br>
<a href="https://github.com/psf/black"><span
  style="font-family:monospace"
>https://github.com/psf/black</span></a>,<br>
<a href="https://black.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://black.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://pypi.org/project/black/"><span
  style="font-family:monospace"
>https://pypi.org/project/black/</span></a>.
</p>

<p>
isort can be found at the following URLs:<br>
<a href="https://pycqa.github.io/isort/"><span
  style="font-family:monospace"
>https://pycqa.github.io/isort/</span></a>,<br>
<a href="https://github.com/pycqa/isort/"><span
```

```
  style="font-family:monospace"
>https://github.com/pycqa/isort/</span></a>,<br>
<a href="https://pypi.org/project/isort/"><span
  style="font-family:monospace"
>https://pypi.org/project/isort/</span></a>.
</p>

<p>
mypy can be found at the following URLs:<br>
<a href="https://mypy-lang.org/"><span
  style="font-family:monospace"
>https://mypy-lang.org/</span></a>,<br>
<a href="https://mypy.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://mypy.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/python/mypy"><span
  style="font-family:monospace"
>https://github.com/python/mypy</span></a>,<br>
<a href="https://pypi.org/project/mypy/"><span
  style="font-family:monospace"
>https://pypi.org/project/mypy/</span></a>.
</p>

<p>
ruff can be found at the following URLs:<br>
<a href="https://docs.astral.sh/ruff/"><span
  style="font-family:monospace"
>https://docs.astral.sh/ruff/</span></a>,<br>
<a href="https://github.com/astral-sh/ruff"><span
  style="font-family:monospace"
>https://github.com/astral-sh/ruff</span></a>,<br>
<a href="https://pypi.org/project/ruff/"><span
  style="font-family:monospace"
>https://pypi.org/project/ruff/</span></a>.
</p>

<p>
bandit can be found at the following URLs:<br>
<a href="https://bandit.readthedocs.io/en/latest/"><span
  style="font-family:monospace"
>https://bandit.readthedocs.io/en/latest/</span></a>,<br>
<a href="https://github.com/PyCQA/bandit"><span
  style="font-family:monospace"
>https://github.com/PyCQA/bandit</span></a>,<br>
<a href="https://pypi.org/project/bandit/"><span
  style="font-family:monospace"
```

```
>https://pypi.org/project/bandit/</span></a>.
</p>

<p>
phpmd can be found at the following URLs:<br>
<a href="https://phpmd.org/"><span
  style="font-family:monospace"
>https://phpmd.org/</span></a>,<br>
<a href="https://github.com/phpmd/phpmd"><span
  style="font-family:monospace"
>https://github.com/phpmd/phpmd</span></a>,<br>
<a href="https://packagist.org/packages/phpmd/phpmd"><span
  style="font-family:monospace"
>https://packagist.org/packages/phpmd/phpmd</span></a>.
</p>

<p>
php can be found at the following URLs:<br>
<a href="https://www.php.net/"><span
  style="font-family:monospace"
>https://www.php.net/</span></a>,<br>
<a href="https://github.com/php/php-src"><span
  style="font-family:monospace"
>https://github.com/php/php-src</span></a>.
</p>

<p>
tex can be found at the following URL:<br>
<a href="https://ctan.org/tex-archive/systems/knuth/dist/tex"><span
  style="font-family:monospace"
>https://ctan.org/tex-archive/systems/knuth/dist/tex</span></a>.
</p>

<p>
pcregrep, pcre2grep can be found at the following URLs:<br>
<a href="https://www.pcre.org/"><span
  style="font-family:monospace"
>https://www.pcre.org/</span></a>,<br>
<a href="https://pcre2project.github.io/pcre2/"><span
  style="font-family:monospace"
>https://pcre2project.github.io/pcre2/</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre1"><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre1</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre2"><span
  style="font-family:monospace"
```

```
>https://github.com/PCRE2Project/pcre2</span></a>,<br>
<a
href="https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c</span
></a>.
</p>

<p>
If you see something that I missed regarding my dependencies,
please tell me/email me :).
</p>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

2.36 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
```

```
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

```

```
% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

```

```
\section{Listing of files}
\label{section:listing}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:\newline
\url{https://www.gnu.org/software/bash/},\newline
```

```
\url{https://tiswww.case.edu/php/chet/bash/bashtop.html},\newline
\url{https://savannah.gnu.org/projects/bash/}.

GNU coreutils can be found at the following URLs:\newline
\url{https://www.gnu.org/software/coreutils/},\newline
\url{https://savannah.gnu.org/projects/coreutils/}.

uutils coreutils can be found at the following URLs:\newline
\url{https://uutils.github.io/coreutils/},\newline
\url{https://github.com/uutils/coreutils}.

diff (GNU diffutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/diffutils/},\newline
\url{https://savannah.gnu.org/projects/diffutils/}.

find, xargs (GNU findutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/findutils/},\newline
\url{https://savannah.gnu.org/projects/findutils/}.

grep (GNU grep) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/grep/},\newline
\url{https://savannah.gnu.org/projects/grep/}.

sed (GNU sed) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/sed/},\newline
\url{https://savannah.gnu.org/projects/sed/}.

ShellCheck can be found at the following URLs:\newline
\url{https://www.shellcheck.net/},\newline
\url{https://github.com/koalaman/shellcheck}.

GNU Wget can be found at the following URLs:\newline
\url{https://www.gnu.org/software/wget/},\newline
\url{https://savannah.gnu.org/projects/wget/}.

more of util-linux can be found at the following URLs:\newline
\url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
},
\newline
\url{https://github.com/util-linux/util-linux} official?

pandoc can be found at the following URLs:\newline
\url{https://pandoc.org/},\newline
\url{https://github.com/jgm/pandoc}.
```

```
pdftex/pdflatex can be found at the following URLs:\newline
\url{https://www.tug.org/applications/pdftex/},\newline
\url{https://tug.org/texlive/svn/}.

tree can be found at the following URLs:\newline
\url{https://oldmanprogrammer.net/source.php?dir=projects/tree},\newline
\url{https://gitlab.com/OldManProgrammer/unix-tree},\newline
\url{https://github.com/Old-Man-Programmer/tree}.

git can be found at the following URLs:\newline
\url{https://git-scm.com/about},\newline
\url{https://github.com/git/git}.

pylint can be found at the following URLs:\newline
\url{https://www.pylint.org/},\newline
\url{https://pylint.readthedocs.io/en/stable/},\newline
\url{https://github.com/pylint-dev/pylint},\newline
\url{https://pypi.org/project/pylint/}.

notify-send can be found at the following URLs:\newline
\url{https://gitlab.gnome.org/GNOME/libnotify/},\newline
\url{https://gnome.pages.gitlab.gnome.org/libnotify/}.

black can be found at the following URLs:\newline
\url{https://github.com/psf/black},\newline
\url{https://black.readthedocs.io/en/stable/},\newline
\url{https://pypi.org/project/black/}.

isort can be found at the following URLs:\newline
\url{https://pycqa.github.io/isort/},\newline
\url{https://github.com/pycqa/isort/},\newline
\url{https://pypi.org/project/isort/}.

mypy can be found at the following URLs:\newline
\url{https://mypy-lang.org/},\newline
\url{https://mypy.readthedocs.io/en/stable/},\newline
\url{https://github.com/python/mypy},\newline
\url{https://pypi.org/project/mypy/}.

ruff can be found at the following URLs:\newline
\url{https://docs.astral.sh/ruff/},\newline
\url{https://github.com/astral-sh/ruff},\newline
\url{https://pypi.org/project/ruff/}.

bandit can be found at the following URLs:\newline
```

```
\url{https://bandit.readthedocs.io/en/latest/},\newline
\url{https://github.com/PyCQA/bandit},\newline
\url{https://pypi.org/project/bandit/}.

phpmd can be found at the following URLs:\newline
\url{https://phpmd.org/},\newline
\url{https://github.com/phpmd/phpmd},\newline
\url{https://packagist.org/packages/phpmd/phpmd}.

php can be found at the following URLs:\newline
\url{https://www.php.net/},\newline
\url{https://github.com/php/php-src}.

tex can be found at the following URL:\newline
\url{https://ctan.org/tex-archive/systems/knuth/dist/tex}.

pcregrep, pcre2grep can be found at the following URLs:\newline
\url{https://www.pcre.org/},\newline
\url{https://pcre2project.github.io/pcre2/},\newline
\url{https://github.com/PCRE2Project/pcre1},\newline
\url{https://github.com/PCRE2Project/pcre2},\newline
{\small
  \url{https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c}
}.

If you see something that I missed regarding my dependencies,
please tell me/email me :).

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

2.37 build_and_checks_variables/files_listing.tex.tpl

```
\subsection{
  build\_and\_checks\_dependencies/build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
}
```

```
\subsection{
  build\_and\_checks\_dependencies/build\_md\_from\_printable\_md.exec.sh
}
\label{
  build_and_checks_dependencies:build_md_from_printable_mdexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_beginnings.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_beginningslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_indentation.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_indentationlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/check\_URLs.libr.sh
}
\label{
  build_and_checks_dependencies:check_URLslibrsh
}
```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_URLs.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/common\_options.libr.sh
}
\label{
  build_and_checks_dependencies:common_optionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_options.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/comparisons.libr.sh
}
\label{
```

```
  build_and_checks_dependencies:comparisonslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/comparisons.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/create\_PDF.exec.sh
}
\label{
  build_and_checks_dependencies:create_PDFexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/create_PDF.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/generate\_from\_template.libr.sh
}
\label{
  build_and_checks_dependencies:generate_from_templatelibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/generate_from_template.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
get\_common\_text\_glob\_patterns.libr.sh
}
\label{
  build_and_checks_dependencies:get_common_text_glob_patternslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
}


\subsection{
```

```
  build\_and\_checks\_dependencies/grammar\_and\_spell\_check.libr.sh
}
\label{
  build_and_checks_dependencies:grammar_and_spell_checklibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
build\_licenses\_templates.exec.sh
}
\label{
  build_and_checks_dependencies:licenses_templates:%
build_licenses_templatesexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
build_licenses_templates.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_AGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_AGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_AGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_GPLv3+.tpl
}
\label{
```

```
  build_and_checks_dependencies:licenses_templates:%
license_file_header_GPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_GPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_LGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_LGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_LGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/lines\_counts.libr.sh
}
\label{
  build_and_checks_dependencies:lines_countslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_counts.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_filters.libr.sh
}
\label{
  build_and_checks_dependencies:lines_filterslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_filters.libr.sh
```

```
}


\subsection{
  build\_and\_checks\_dependencies/lines\_maps.libr.sh
}
\label{
  build_and_checks_dependencies:lines_mapslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_maps.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/overwrite\_if\_not\_equal.libr.sh
}
\label{
  build_and_checks_dependencies:overwrite_if_not_equallibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_black\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_black_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_black_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_isort\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_isort_complementlibrsh
}

```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_isort_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/repository\_data.libr.sh
}
\label{
  build_and_checks_dependencies:repository_datalibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/repository_data.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.exec.php
}
\label{
  build_and_checks_dependencies:split_scoreexecphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.exec.php
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.libr.php
}
\label{
  build_and_checks_dependencies:split_scorelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.libr.php
}


\subsection{
  build\_and\_checks\_dependencies/strings\_functions.libr.sh
}
\label{
  build_and_checks_dependencies:strings_functionslibrsh
```

```
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/strings_functions.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/too\_long\_code\_lines.libr.sh
}
\label{
  build_and_checks_dependencies:too_long_code_lineslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/too_long_code_lines.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:update_common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_or\_check\_files\_names\_listing.exec.sh
}
\label{
  build_and_checks_dependencies:update_or_check_files_names_listingexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh
}


\subsection{
```

```
  build\_and\_checks\_variables/.gitignore
}
\label{
  build_and_checks_variables:gitignore
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/.gitignore
}


\subsection{
  build\_and\_checks\_variables/bandit.ini
}
\label{
  build_and_checks_variables:banditini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit.ini
}


\subsection{
  build\_and\_checks\_variables/bandit\_baseline.json
}
\label{
  build_and_checks_variables:bandit_baselinejson
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit_baseline.json
}


\subsection{
  build\_and\_checks\_variables/black.toml
}
\label{
  build_and_checks_variables:blacktoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/black.toml
}

```

```
\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.html.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptshtmltpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.tex.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptstextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_listing.tex.tpl
}
\label{
  build_and_checks_variables:files_listingtextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_listing.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_names\_listing.txt
}
\label{
  build_and_checks_variables:files_names_listingtxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_names_listing.txt
```

```
}


\subsection{
  build\_and\_checks\_variables/licenses\_notes.txt
}
\label{
  build_and_checks_variables:licenses_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/licenses_notes.txt
}


\subsection{
  build\_and\_checks\_variables/phpmd\_baseline.xml
}
\label{
  build_and_checks_variables:phpmd_baselinexml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/phpmd_baseline.xml
}


\subsection{
  build\_and\_checks\_variables/pylintrc.toml
}
\label{
  build_and_checks_variables:pylintrctoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/pylintrc.toml
}


\subsection{
  build\_and\_checks\_variables/repository\_data.txt
}
\label{
  build_and_checks_variables:repository_datatxt
}

```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/repository_data.txt
}


\subsection{
  build\_and\_checks\_variables/ruff.toml
}
\label{
  build_and_checks_variables:rufftoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/ruff.toml
}


\subsection{
  build\_and\_checks\_variables/shellcheck.ini
}
\label{
  build_and_checks_variables:shellcheckini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck.ini
}


\subsection{
  build\_and\_checks\_variables/shellcheck\_notes.txt
}
\label{
  build_and_checks_variables:shellcheck_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck_notes.txt
}


\subsection{
  build\_and\_checks.exec.sh
}
\label{
  build_and_checksexecsh
```

```
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks.exec.sh
}


\subsection{
  COPYING
}
\label{
  COPYING
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING
}


\subsection{
  COPYING.LESSER
}
\label{
  COPYINGLESSER
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING.LESSER
}


\subsection{
  git\_helpers/git\_checks.libr.sh
}
\label{
  git_helpers:git_checkslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/git_checks.libr.sh
}


\subsection{
  git\_helpers/post-commit
}
```

```
\label{
  git_helpers:post-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post-commit
}


\subsection{
  git\_helpers/post\_commit\_renaming\_comment.libr.sh
}
\label{
  git_helpers:post_commit_renaming_commentlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post_commit_renaming_comment.libr.sh
}


\subsection{
  git\_helpers/pre-commit
}
\label{
  git_helpers:pre-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/pre-commit
}


\subsection{
  IMAP\_transfert.py
}
\label{
  IMAP_transfertpy
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
IMAP_transfert.py
}


\subsection{
```

```
  LiensUtiles.txt
}
\label{
  LiensUtilestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
LiensUtiles.txt
}


\subsection{
  miscellaneous/spampty\_data.txt
}
\label{
  miscellaneous:spampty_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/spampty_data.txt
}


\subsection{
  README.md.tpl
}
\label{
  READMEmdtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
README.md.tpl
}


\subsection{
  unicode.libr.php
}
\label{
  unicodelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
unicode.libr.php
}

```

```

\subsection{
  upgrade\_build\_and\_checks.exec.sh
}
\label{
  upgrade_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
upgrade_build_and_checks.exec.sh
}


\subsection{
  wget\_sha512.libr.sh
}
\label{
  wget_sha512librsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
wget_sha512.libr.sh
}


```

2.38 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
```

```
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
```

```
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_listing.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
miscellaneous/spampty_data.txt
// README.html
// README.md
README.md.tpl
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

2.39 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

```

```
Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
```

```
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/
```

```
notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
```

```
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

2.40 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName" file="build_and
  <violation rule="PHPMD\Rule\Naming\LongVariable" file="build_and_checks_dependenc
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName" file="build_and_
  <violation rule="PHPMD\Rule\Design\LongMethod" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName" file="unicode.l
  <violation rule="PHPMD\Rule\Naming\LongVariable" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName" file="unicode.li
  <violation rule="PHPMD\Rule\CleanCode\MissingImport" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\CyclomaticComplexity" file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Design\NpathComplexity" file="unicode.libr.php"/>
</phpmd-baseline>
```

2.41 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
```

```
[tool.pylint.reports]
output-format = "colorized"
```

2.42 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

grammar_or_spell_checker_command=more


```

```
Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
        "0.1.0": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "4.4.4" : {}
          }
        }
      }
    }
  },
  "B": {
```

```
    "2.2.2": {
      "C": {
        // Dumb, you did not found it by yourself? Failed exam.
        "4.4.4" : {}
      }
    }
  }
}
Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.

bandit_venv=../venv_bandit
black_venv=../venv_black
isort_venv=../venv_isort
mypy_venv=../venv_mypy
pylint_venv=../venv_pylint
ruff_venv=../venv_ruff


upgrade venvs can be auto or ask, any other value is interpreted
as ask.

upgrade_venvs=ask


time interval can be 0 to always upgrade or ask to upgrade,
or a number at most bash max int
https://superuser.com/questions/1030122/
what-is-the-maximum-value-of-a-numeric-bash-shell-variable
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
```

```
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800
```

2.43 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
```

```
]
```

2.44 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"

```

2.45 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck:
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
           False positive when c contains \n.
- SC2034 : Unused variable
           False positive for globals that are declared in functions
           that are not currently called.
           Somehow it can be used to detect dead code.
- SC2086 : Double quote to prevent globbing and word splitting.
           False positive when returning integer variable
           via an indirection like return ${!foo}.
           See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
           False positive when passing arguments would yield
           a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
           False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
           False positive when assigning a variable declared with -i.
           https://github.com/koalaman/shellcheck/issues/914
```

```
- SC2145 : Argument mixes string and array.
           False positive when using "$@" line return and pipes.
           https://github.com/koalaman/shellcheck/issues/3192
           Example:
             # shellcheck disable=SC2145
             all_self_code_lines "$@"\
               | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
           Very useful but false positive when I play with
           variable variable names.
- SC2162 : read without -r will mangle backslashes.
           False positive when read is used without variable:
           https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
           False positive on returning integer values.
           https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
             Use = to compare as string (or expand explicitly with $var).
           Very useful but false positive when I play with
           variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
           value is masked. Frequent with pipes.

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2086$" -e "=SC2119$" -e "=SC2120$" -e "=SC2125$"\
  -e "=SC2145$" -e "=SC2154$" -e "=SC2162$" -e "=SC2248$" -e "=SC2309$"\
  -e "=SC2312$" -e "=SC2154,SC2309$" -e "=SC2162,SC2312$"
```

2.46 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    {
      echo "temp/"
      echo ""
    } > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
```

```
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='4696142346f9a90b479491908301e460cdd44093fe49b'
  LFBFL_correct_sha512+='bbe5de6075ac9ef4e00510a856c705143833b3d23f7d'
  LFBFL_correct_sha512+='9aeb4fae8d58255945f12707661df32971c406b'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed_point_build* ]]; then
  echo "--fixed_point_build"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
```

```
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed_point_build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.47 COPYING

```
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```
<one line to give the program's name and a brief idea of what it does.>
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```

```

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Application with the Library.  The particular version of the Library
with which the Combined Work was made is also called the "Linked
Version".

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Corresponding Source for the Combined Work, excluding any source code
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object code and/or source code for the Application, including any data
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  If you modify a copy of the Library, and, in your modifications, a
facility refers to a function or data to be supplied by an Application
```

```
that uses the facility (other than as an argument passed when the
facility is invoked), then you may convey a copy of the modified
version:

   a) under this License, provided that you make a good faith effort to
   ensure that, in the event an Application does not supply the
   function or data, the facility still operates, and performs
   whatever part of its purpose remains meaningful, or

   b) under the GNU GPL, with none of the additional permissions of
   this License applicable to that copy.

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  The object code form of an Application may incorporate material from
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   the Library is used in it and that the Library and its use are
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   document.

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   execution, include the copyright notice for the Library among
   these notices, as well as a reference directing the user to the
```

```
    copies of the GNU GPL and this license document.

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        manner specified by section 6 of the GNU GPL for conveying
        Corresponding Source.

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        Library.  A suitable mechanism is one that (a) uses at run time
        a copy of the Library already present on the user's computer
        system, and (b) will operate properly with a modified version
        of the Library that is interface-compatible with the Linked
        Version.

    e) Provide Installation Information, but only if you would otherwise
    be required to provide such information under section 6 of the
    GNU GPL, and only to the extent that such information is
    necessary to install and execute a modified version of the
    Combined Work produced by recombining or relinking the
    Application with a modified version of the Linked Version. (If
    you use option 4d0, the Installation Information must accompany
    the Minimal Corresponding Source and Corresponding Application
    Code. If you use option 4d1, you must provide the Installation
    Information in the manner specified by section 6 of the GNU GPL
    for conveying Corresponding Source.)

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facilities that are not Applications and are not covered by this
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choice, if you do both of the following:

    a) Accompany the combined library with a copy of the same work based
    on the Library, uncombined with any other library facilities,
    conveyed under the terms of this License.

    b) Give prominent notice with the combined library that part of it
    is a work based on the Library, and explaining where to find the
    accompanying uncombined form of the same work.
```

```

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apply, that proxy's public statement of acceptance of any version is
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Library.
```

2.49 git_helpers/git_checks.libr.sh

```
#!/usr/bin/env bash
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```

```
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#
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# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle"
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits -> #Prefetch()"
  declare -i LFBFL_error=0
  git diff --cached --name-only\
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
    LFBFL_error=1
  fi
  if grep serializer /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
    LFBFL_error=1
  fi
```

```
  rm /tmp/DOSAS_django_git_check1.temp
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}


check_no_abusive_trailing_comma(){
  # Useful for Python code, although it regularly yields
  # false positives in Python code
  # (but it is easy to distinguish by hand).
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION"
  local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
  LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
  readonly LFBFL_send_body_1
  declare -i LFBFL_error=0
  git diff --cached -r\
    > /tmp/DOSAS_django_git_check2.temp
  if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    LFBFL_error=1
  fi
  rm /tmp/DOSAS_django_git_check2.temp
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_files_string=$(
    git diff --cached --name-only\
    | grep '\.py'
  )
  echo "${LFBFL_files_string}"
  declare -a LFBFL_some_files
  mapfile -t LFBFL_some_files <<< "${LFBFL_files_string}"
  readonly LFBFL_some_files
  declare -i LFBFL_error=0
  # Not very useful since LFBFL_file should be local to the loop by
  # default.
  local LFBFL_file
  for LFBFL_file in "${LFBFL_some_files[@]}"; do
    echo "Black will check formatting on file ${LFBFL_file}"
    if ! black --check --diff "${LFBFL_file}";
```

```
    then
      LFBFL_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}
```

2.50 git_helpers/post-commit

```
#!/usr/bin/env bash
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#
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# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="post_commit_renaming_comment.libr.sh"
# shellcheck source=post_commit_renaming_comment.libr.sh
source "${LFBFL_library}"
```

```
LFBFL_library="../DevOrSysAdminScripts/build_and_checks_dependencies/"
LFBFL_library+="lines_filters.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "${LFBFL_library}"

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

2.51 git_helpers/post_commit_renaming_comment.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
```

```
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_cd_result2
pushd .
cd "${LFBFL_some_directory}" || {
  LFBFL_cd_result2=$?
  echo "post_commit_renaming_comment.libr.sh no such directory"
  # shellcheck disable=SC2248
  exit ${LFBFL_cd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_popd_result2
popd || {
  LFBFL_popd_result2=$?
  echo "post_commit_renaming_comment.libr.sh no popd"
  # shellcheck disable=SC2248
  exit ${LFBFL_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
```

```
72    declare -i LFBFL_i_verbose=0
73    get_verbose_option "$@"
74    local LFBFL_work_directory=""
75    get_work_directory_option "$@"
76    local LFBFL_arg
77
78    declare -i LFBFL_log_directory_change=0
79    if [[ "$*" == *--log-directory-change* ]]; then
80      LFBFL_log_directory_change=1
81    fi
82    readonly LFBFL_log_directory_change
83
84    declare -i LFBFL_max_comment_line_length=70
85    for LFBFL_arg in "$@"; do
86      if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
87        LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
88        LFBFL_max_comment_line_length=$((LFBFL_arg))
89        break
90      fi
91    done
92    readonly LFBFL_max_comment_line_length
93
94    if [[ LFBFL_i_verbose -eq 1 ]]; then
95      echo "LFBFL_log_directory_change: ${LFBFL_log_directory_change}"
96      echo "LFBFL_max_comment_line_length:"\
97        "${LFBFL_max_comment_line_length}"
98    fi
99    #-----------------------------------------------------------------------
100
101   enhanced_set_shell_option pipefail\
102     && trap 'enhanced_unset_shell_option pipefail' RETURN
103
104   declare -r LFBFL_renaming_lines=$(
105     git log -p -1\
106     | grep '^diff --git' -A 1
107   )
108   if [[ LFBFL_i_verbose -eq 1 ]]; then
109     echo "LFBFL_renaming_lines: ${LFBFL_renaming_lines}"
110   fi
111
112   declare -a LFBFL_renaming_lines_array
113   mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
114   readonly LFBFL_renaming_lines_array
115
116   declare -r LFBFL_timestamp=$(date --iso-8601="minute")
117   if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
118       echo "LFBFL_timestamp: ${LFBFL_timestamp}"
119     fi
120     get_split_score_simple 1 "${LFBFL_max_comment_line_length}" _
121     LFBFL_split_score_command1="${get_split_score_result}"
122     LFBFL_split_score_command_properties1="${get_split_score_result2}"
123     get_split_score_simple 1 "${LFBFL_max_comment_line_length}" /
124     LFBFL_split_score_command2="${get_split_score_result}"
125     LFBFL_split_score_command_properties2="${get_split_score_result2}"
126     declare -i LFBFL_renaming_happened=0
127     declare -i LFBFL_i
128     declare -i LFBFL_j
129     # declare -r LFBFL_special_file_name="files_names_listing.txt"
130     declare -r LFBFL_sfn="files_names_listing.txt"
131     local LFBFL_diff_line
132     local LFBFL_similarity_line
133     local LFBFL_old_file_path
134     local LFBFL_old_file_name
135     local LFBFL_old_file_directory
136     local LFBFL_new_file_path
137     local LFBFL_new_file_name
138     local LFBFL_new_file_directory
139     local LFBFL_useful_file_name
140     local LFBFL_extension
141     local LFBFL_comment_prefix
142     local LFBFL_comment_prefix2
143     local LFBFL_old_file_name2
144     local LFBFL_old_file_directory2
145     local LFBFL_new_file_name2
146     local LFBFL_new_file_directory2
147     local LFBFL_new_comment
148     declare -i LFBFL_copyright_line_number
149     declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
150     declare -i LFBFL_line_count
151     declare -i LFBFL_lines_after
152     local LFBFL_temp_file_path
153     for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
154       LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
155       LFBFL_i=$((LFBFL_i + 1))
156       LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
157       LFBFL_i=$((LFBFL_i + 1))
158       # We need to increment 3 times since grep adds "--" lines between.
159       LFBFL_i=$((LFBFL_i + 1))
160       if [[ LFBFL_i_verbose -eq 1 ]]; then
161         echo "LFBFL_diff_line: ${LFBFL_diff_line}"
162         echo "LFBFL_similarity_line: ${LFBFL_similarity_line}"
163       fi
```

```
    if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
      continue
    fi

    # Extract old file name from diff line. --------------------------
    LFBFL_old_file_path=$(
      echo "${LFBFL_diff_line}"\
      | cut -d ' ' -f 3
    )
    # It starts by "a/".
    LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
    LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
    LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
    # Extract new file name from diff line. --------------------------
    LFBFL_new_file_path=$(
      echo "${LFBFL_diff_line}"\
      | cut -d ' ' -f 4
    )
    # It starts by "b/".
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      echo "LFBFL_old_file_path: ${LFBFL_old_file_path}"
      echo "LFBFL_old_file_name: ${LFBFL_old_file_name}"
      echo "LFBFL_old_file_directory: ${LFBFL_old_file_directory}"
      echo "LFBFL_new_file_path: ${LFBFL_new_file_path}"
      echo "LFBFL_new_file_name: ${LFBFL_new_file_name}"
      echo "LFBFL_new_file_directory: ${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. ------------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      echo "LFBFL_useful_file_name: ${LFBFL_useful_file_name}"
      echo "LFBFL_extension: ${LFBFL_extension}"
    fi
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        echo "LFBFL_useful_file_name: ${LFBFL_useful_file_name}"
```

```
210         echo "LFBFL_extension: ${LFBFL_extension}"
211       fi
212     done
213     LFBFL_comment_prefix=""
214     if [[ "${LFBFL_extension}" == "sh" ]]; then
215       LFBFL_comment_prefix="# "
216     fi
217     if [[ "${LFBFL_extension}" == "sql" ]]; then
218       LFBFL_comment_prefix="-- "
219     fi
220     if [[ "${LFBFL_extension}" == "tex" ]]; then
221       LFBFL_comment_prefix="% "
222     fi
223     if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
224       LFBFL_comment_prefix="// "
225     fi
226     if [[ LFBFL_i_verbose -eq 1 ]]; then
227       echo "LFBFL_comment_prefix: ${LFBFL_comment_prefix}"
228     fi
229     #-----------------------------------------------------------------
230     # Split if necessary old_file_name, new_file_name, etc. to have
231     # comment lines below max_comment_line_length.
232     LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
233     # -----
234     LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
235     LFBFL_old_file_name2+="${LFBFL_old_file_name}"
236     repeated_split_last_line "${LFBFL_old_file_name2}"\
237       "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
238       '"' "${LFBFL_split_score_command1}"\
239       "${LFBFL_split_score_command_properties1}"
240     LFBFL_old_file_name2="${repeated_split_last_line_result}"
241     # -----
242     LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
243     LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
244     repeated_split_last_line "${LFBFL_old_file_directory2}"\
245       "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
246       '"' "${LFBFL_split_score_command2}"\
247       "${LFBFL_split_score_command_properties2}"
248     LFBFL_old_file_directory2="${repeated_split_last_line_result}"
249     # -----
250     LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
251     LFBFL_new_file_name2+="${LFBFL_new_file_name}"
252     repeated_split_last_line "${LFBFL_new_file_name2}"\
253       "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
254       '"' "${LFBFL_split_score_command1}"\
255       "${LFBFL_split_score_command_properties1}"
```

```
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_split_score_command_properties2}"
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. -----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
    then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from\n"
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to\n"
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_log_directory_change -eq 1 ]] && [[\
      "${LFBFL_new_file_directory}"\
      != "${LFBFL_old_file_directory}"\
    ]]; then
      if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
      then
        LFBFL_new_comment+="\n"
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from\n"
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to\n"
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      echo "LFBFL_new_comment: ${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. -------------------------------------
    LFBFL_copyright_line_number=$(
      grep -n '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut -d ':' -f 1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      echo "LFBFL_copyright_line_number:"\
```

```
302         "${LFBFL_copyright_line_number}"
303       fi
304       if [[ LFBFL_copyright_line_number -eq 0 ]]; then
305         notify-send "${LFBFL_send_summary_2}"\
306           "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
307         if [[ LFBFL_i_verbose -eq 1 ]]; then
308           echo "No copyright line. Skipping file."
309         fi
310         continue
311       fi
312       # Get total number of lines. -------------------------------------
313       LFBFL_line_count=$(
314         wc -l\
315         < "${LFBFL_new_file_path}"
316       )
317       if [[ LFBFL_i_verbose -eq 1 ]]; then
318         echo "LFBFL_line_count: ${LFBFL_line_count}"
319       fi
320       # Compute number of lines after. ---------------------------------
321       LFBFL_lines_after=$((
322         LFBFL_line_count - LFBFL_copyright_line_number
323       ))
324       if [[ LFBFL_i_verbose -eq 1 ]]; then
325         echo "LFBFL_lines_after: ${LFBFL_lines_after}"
326       fi
327       # Update file. ---------------------------------------------------
328       LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
329       {
330         head --lines="${LFBFL_copyright_line_number}"\
331           "${LFBFL_new_file_path}"
332         echo -e "${LFBFL_new_comment}"
333         tail --lines="${LFBFL_lines_after}" "${LFBFL_new_file_path}"
334       } >> "${LFBFL_temp_file_path}"
335       mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
336       # Add file to stage. ---------------------------------------------
337       git add "${LFBFL_new_file_path}"
338       LFBFL_renaming_happened=1
339     done
340     readonly LFBFL_renaming_happened
341     if [[ LFBFL_renaming_happened -eq 1 ]]; then
342       # Commit ---------------------------------------------------------
343       local LFBFL_message="commit_a_file_renamed_comment()"
344       LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
345       readonly LFBFL_message
346       git commit --message="${LFBFL_message}"
347     fi
```

```
}
```

2.52 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# either version 3 of the License,
# or (at your option) any later version.
#
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# but WITHOUT ANY WARRANTY;
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# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="git_checks.libr.sh"
# shellcheck source=git_checks.libr.sh
source "${LFBFL_library}"

declare -ig LFBFL_git_hook_error=0
if ! check_files; then
  LFBFL_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
```

```
  LFBFL_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_git_hook_error=1
fi
readonly LFBFL_git_hook_error
if [[ LFBFL_git_hook_error -eq 1 ]]; then
  exit 1
fi
exit 0
```

2.53 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


```

```
class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
```

```
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
```

```
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
```

```
172             "Transfert all emails from some IMAP account"
173             " to another IMAP account"
174         ),
175     )
176     parser.add_argument(
177         "from_login",
178         type=str,
179         help="Email account login of user one.",
180     )
181     parser.add_argument(
182         "from_password",
183         type=str,
184         help="Email account password of user one.",
185     )
186     parser.add_argument(
187         "to_login",
188         type=str,
189         help="Email account login of user two.",
190     )
191     parser.add_argument(
192         "to_password",
193         type=str,
194         help="Email account password of user two.",
195     )
196     parser.add_argument(
197         "--to_new_folder",
198         type=str,
199         default="",
200         help="Transfert emails into a new folder of this name.",
201     )
202     parser.add_argument(
203         "--from_host",
204         type=str,
205         default="imap.gmail.com",
206         help="Host of first email account",
207     )
208     parser.add_argument(
209         "--from_port",
210         type=int,
211         default=993,
212         help="Port on host of first email account",
213     )
214     parser.add_argument(
215         "--to_host",
216         type=str,
217         default="imap.gmail.com",
```

```
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-------------------------------------------------------\n"
            "IMAP transfert:\n"
            "-------------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-------------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
```

```
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

2.54 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
```

2.55 miscellaneous/spampty_data.txt

```
a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

2.56 README.md.tpl

```
# DevOrSysAdminScripts

[![CodeFactor-badge]][CodeFactor-package-page]
[![Codacy-badge]][Codacy-package-page]
![GitHub-top-language-badge]
![GitHub-code-size-in-bytes-badge]
[![Security:bandit:badge]][Bandit-GitHub.com]
[![Code-style:black:badge]][Black-GitHub.com]
[![Imports:isort:badge]][Isort-GitHub.io]
[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
```

```
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.

But the files "pre-commit" and "post-commit" were not renamed,
since it is not possible to give them other names to use them as
pre-commit/post-commit hooks in `git`.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

[Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

[GitHub-top-language-badge]: https://img.shields.io/github/\
languages/top/llyaudet/DevOrSysAdminScripts

[GitHub-code-size-in-bytes-badge]: https://img.shields.io/github/\
languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/
```

```

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

2.57 unicode.libr.php

```
<?php
/*
This file is part of DevOrSysAdminScripts library.

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This file was renamed from "unicode.php" to "unicode.libr.php".
*/

function decimal_code_point_to_UTF8(
  $i_code_point_in_decimal_notation
){
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
```

```
33    if($i_code_point_in_decimal_notation < 128){
34      return chr($i_code_point_in_decimal_notation);
35    }
36    $i_continuation_base_value = 128;
37    // 110xxxxx 10xxxxxx
38    if($i_code_point_in_decimal_notation < 256 * (2**3)){
39      $i_first_byte_base_value = 192;
40      $i_last_byte_significant_bits = (
41        $i_code_point_in_decimal_notation % (2**6)
42      );
43      $i_first_byte_significant_bits = intdiv(
44        (
45          $i_code_point_in_decimal_notation
46          - $i_last_byte_significant_bits
47        ),
48        2**6,
49      );
50      $s_result = chr(
51        $i_first_byte_base_value
52        + $i_first_byte_significant_bits
53      );
54      $s_result .= chr(
55        $i_continuation_base_value
56        + $i_last_byte_significant_bits
57      );
58      return $s_result;
59    }
60    // 1110xxxx 10xxxxxx 10xxxxxx
61    if($i_code_point_in_decimal_notation < 256 * 256){
62      $i_first_byte_base_value = 224;
63      $arr_arr_data_per_byte_reverse = [
64        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
65        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
66        ["i_bits" => 4, "i_base_value" => $i_first_byte_base_value],
67      ];
68      $i_rest = $i_code_point_in_decimal_notation;
69      $s_result = "";
70      foreach($arr_arr_data_per_byte_reverse as $arr_data){
71        // _Pragma unroll
72        // J'aurais pu "roller" les 4 cas des ifs.
73        // Mais pour les perfs, il faudrait tout unroller vu le nombre
74        // d'itérations. Et j'aimerais bien comparer du code unrollé
75        // aux 2 niveaux à la main avec celui unrollé par le compilateur
76        // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
77        // l'intérieur d'abord (pour ne pas refaire le même boulot)
78        // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
```

```
      // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
      $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data["i_bits"],
      );
      $s_result = (  // .= or =.
        chr($i_significant_bits + $arr_data["i_base_value"])
        .$s_result
      );
    }
    return $s_result;
  }
  // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 256 * 256 * (2**5)){
    $i_first_byte_base_value = 240;
    $arr_arr_data_per_byte_reverse = [
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 3, "i_base_value" => $i_first_byte_base_value],
    ];
    $i_rest = $i_code_point_in_decimal_notation;
    $s_result = "";
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data["i_bits"],
      );
      $s_result = (  // .= or =.
        chr($i_significant_bits + $arr_data["i_base_value"])
        .$s_result
      );
    }
    return $s_result;
  }
  throw new Exception(
    "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
    ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
    ." 3 octets au lieu de 4 :face-palm:."
    ." Du coup, ça peut donner quelques \"surprises\"."
    ." Sinon, y'a aussi ASCII dans MySQL."
    ." Et comme ça joue sur la mémoire et les performances,"
    ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
```

```
125      ." :)"
126    );
127  }
128
129
130
131  function hexa_code_point_to_UTF8(
132    $s_code_point_in_hexadecimal_notation
133  ){
134    $s_code_point_in_hexadecimal_notation = strtoupper(
135      $s_code_point_in_hexadecimal_notation
136    );
137    $i_code_point_in_decimal_notation = 0;
138    $i_ord_0 = ord('0');
139    $i_ord_A = ord('A');
140    $i_digit_offset = -$i_ord_0;
141    $i_letter_offset = 10 - $i_ord_A;
142    for(
143      $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
144      $i < $i_max;
145      ++$i
146    ){
147      $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
148      $i_code_point_in_decimal_notation *= 16;
149      if(ctype_digit($s_hexa_digit)){
150        $i_code_point_in_decimal_notation += (
151          ord($s_hexa_digit) + $i_digit_offset
152        );
153        continue;
154      }
155      if(ctype_xdigit($s_hexa_digit)){
156        $i_code_point_in_decimal_notation += (
157          ord($s_hexa_digit) + $i_letter_offset
158        );
159        continue;
160      }
161      throw new Exception("Not an hexadecimal digit.");
162    }
163    return decimal_code_point_to_UTF8(
164      $i_code_point_in_decimal_notation
165    );
166  }
167
168
169
170  function check_string_is_valid_ASCII($s_string){
```

```
171   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
172     if(ord($s_string[$i]) > 127){
173       throw new Exception(
174         "Non-ASCII character found at octet position "
175         .$i
176         ., value "
177         .ord($s_string[$i])
178         ."."
179       );
180     }
181   }
182   return true;
183 }
184
185
186
187 function check_file_is_valid_ASCII($s_file_name){
188   $s_string = file_get_contents($s_file_name);
189   if($s_string === false){
190     throw new Exception("File ".$s_file_name." not found.");
191   }
192   return check_string_is_valid_ASCII($s_string);
193 }
194
195
196
197 function check_string_is_valid_UTF8($s_string){
198   $i_continuation_octet_needed = 0;
199   $i_character_start_position = 0;
200   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
201     $i_current_octet = ord($s_string[$i]);
202     if($i_continuation_octet_needed > 0){
203       if($i_current_octet < 128 || $i_current_octet >= 192){
204         throw new Exception(
205           "Non-UTF8 character found at start position "
206           .$i_character_start_position
207           ., octet at position "
208           .$i
209           ." has value "
210           .$i_current_octet
211           ." which is not a continuation octet."
212         );
213       }
214       --$i_continuation_octet_needed;
215       continue;
216     }
```

```
217
218       $i_character_start_position = $i;
219       if($i_current_octet < 128){ // 0xxxxxxx ASCII
220         continue;
221       }
222       if($i_current_octet >= 128 && $i_current_octet < 192){
223         throw new Exception(
224           "Non-UTF8 character found at start position "
225           .$i_character_start_position
226           ., octet at position "
227           .$i
228           ." has value "
229           .$i_current_octet
230           ." which is a continuation octet."
231         );
232       }
233       if($i_current_octet >= 192 && $i_current_octet < 224){
234         // 110xxxxx 10xxxxxx
235         $i_continuation_octet_needed = 1;
236         continue;
237       }
238       if($i_current_octet >= 224 && $i_current_octet < 240){
239         // 1110xxxx 10xxxxxx 10xxxxxx
240         $i_continuation_octet_needed = 2;
241         continue;
242       }
243       if($i_current_octet >= 240 && $i_current_octet < 248){
244         // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
245         $i_continuation_octet_needed = 3;
246         continue;
247       }
248       throw new Exception(
249         "Non-UTF8 character found at start position "
250         .$i_character_start_position
251         ., octet at position "
252         .$i
253         ." has value "
254         .$i_current_octet
255         ." which is invalid."
256       );
257     }
258     return true;
259   }
260
261
262
```

```
function check_file_is_valid_UTF8($s_file_name){
  $s_string = file_get_contents($s_file_name);
  if($s_string === false){
    throw new Exception("File ".$s_file_name." not found.");
  }
  return check_string_is_valid_UTF8($s_string);
}



/*
?>
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
```

```
}
//*/

```

2.58 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 $*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed_point_build* ]]; then
    LFBFL_fixed_point_build="--fixed_point_build"
  fi
  readonly LFBFL_verbose
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
```

```
  )
  ./build_and_checks.exec.sh "." "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"
}

upgrade_build_and_checks "$@"
```

2.59 wget_sha512.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1 file_path
  # $2 download_URL
  # $3 correct_sha512
  # $4 optional --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    echo "$0 wget_sha512 $*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
```

```
  if [[ ! -f "$1" ]];
  then
    wget "${LFBFL_verbose}" -O "$1" "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum "$1"\
    | cut -f1 -d' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]];
  then
    echo "$1 does not have correct sha512"
    echo "wanted $3"
    echo "found ${LFBFL_present_sha512}"
    exit
  fi
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !